



FLIR METERLiNK 4.2

METERLiNK 4.2

Specification

METERLiNK 4.2

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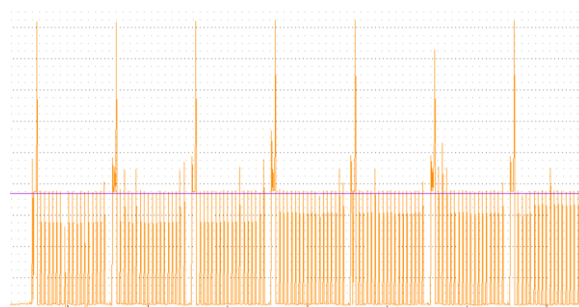




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Revision History

Revision	Date	Author	Comment
0.1	11/23/2020	Henry Li	Draft version
0.2	02/24/2021	Sylar Lan	- Change Sensor Reading package format - Add Meter Type index
0.3	03/02/2021	Sylar Lan	- Add unit index for Fc and Lux - Add Sensor reading protocol structure
0.4	03/10/2021	Sylar Lan	Update Metertype, Function and Unit index character for application UI display.
0.5	03/15/2021	Sylar Lan	- Add Fast/Low in meter status byte - Add Auto Power off in battery status - Add Subunits index into sensor reading
0.6	04/05/2021	Sylar Lan	Update UUID of Sensor Reading and DFU service
0.7	04/26/2021	Sylar Lan	Change Fast/Slow sampling rate indicator to Meter Status Flag2 (Byte20)
1.1	06/28/2022	Sylar Lan	Initial Bulk data transferring spec.
4.1.2	07/11/2022	Sylar Lan	Update bulk data transferring spec
4.1.62	07/25/2022	Sylar Lan	- Update METERLiNK Tx Command - Update METERLiNK Rx Command
4.1.63	08/03/2022	Sylar Lan	Update Bulk Data Transferring Error Handling process
4.1.7	08/05/2022	Sylar Lan	- Update Bulk Data single file error handling process - Update Bulk Data multiple files handling process - Update Bulk Data File Transferring process
4.1.71	08/10/2022	Sylar Lan	Update log file format
4.1.72	08/31/2022	Sylar Lan	- Add function log display name - Add example of One-shot log (0x04) and dynamic log (0x05) regarding Bulk data's Data field
4.1.73	09/27/2022	Sylar Lan	- Remove Bulk transferring "one-shot log" file type - Fixed the typo of BULK_INFO_REPLY sub op code - Change BULK_INFO Total packet no. definition - Update ML_APP_MSG_CONTROL_REPLY Cmd - Update bulk data transferring process - Update log format
4.1.74	10/19/2022	Sylar Lan	Add "event to app" in "Sensor Reading"
4.1.75	11/01/2022	Sylar Lan	- Update more detail about Peak, Max Min and Hold flag - Update more detail about RMS and OL flag - Update example of Event to App - Update Function Name Index - Update App OTA cmd to Meter
4.1.76	01/10/2023	Sylar Lan	- Update Log raw file format note - Update RSSI Weak Signal Notification - Update OL flag should be filling "Value", "Digits to display" and "Metric Prefix"
4.1.77	01/31/2023	Sylar Lan	Add Watt AC DC Volt and Current function index
4.1.78	02/14/2023	Sylar Lan	- Add "Data description" filed in sensor reading packet - Modify "display app name" in function index



			• Add 0x02 into “event to App” to send Last reading data
4.1.79	02/23/2023	Sylar Lan	• Update function index table “Display in app” name • Add ML_DEV_ENG_PACKEET_LOST_TEST cmd • Update Packet length in Bulk info and Bulk Data command with MTU information • Remove 0x90 (power factor) function index
4.1.80	06/01/2023	Sylar Lan	• Add Lepton related function index and unit index
4.1.81	06/29/2023	Sylar Lan	• Add “Greater than value” bit and “Less than value” bit in “Meter Status Flag2” • Add “Voltage Warning” bit in “Meter status Flag1” • Add “Insulation” function and PI, DAR, CMP index number • Add sub index into Ohm index
4.1.82	07/29/2023	Sylar Lan	• Add “Data Description” definition • Remove “COMP Ohm” from “subunit index” and move to “Data Description”
4.1.82.1	08/01/2023	Sylar Lan	• Update “Function” and “Unit” index tables
4.1.82.3	09/09/2023	Sylar Lan	• Add Period into Function index
4.1.82.4	09/23/2023	Sylar Lan	• Add Decibel (dB) into Function index • Add AC+DC Volt. LoZ into Function index
4.1.82.6	10/10/2023	Sylar Lan	• Add PV DC A function index
4.1.82.7	10/16/2023	Sylar Lan	• Modify Photovoltaic DC 0x8E function index
4.1.84	06/20/2024	Sylar Lan	• Add low battery code in app response code
4.1.86	11/14/2024	Sylar Lan	• Add new protocol id in device info • Add EVSE Tester process into Data Description • Add 0x06 in ML_DEV_BULK_INFO for EVST Tester
4.1.89	12/12/2024	Sylar Lan	• Add Function index for Inrush
4.1.90	03/25/2025	Sylar Lan	• OL The unit will follow Metric Prefix
4.1.91	07/03/2025	Sylar Lan	• Add Meter type for EVSE Tester 0x18s
4.1.92	11/17/2025	Sylar Lan	• Correct typo
4.2.1	12/12/2025	Sylar Lan	• Add EVSE meter ID to 0xF0 • Add Vibration sensor ID to 0xE0 and ISO table • Add Vibration sensor function name from 0x97 to 9xA0 • Modify display name for function index 0x6C to 0x6F
4.2.11	12/24/2025	Sylar Lan	• Add “Enable Slave Y = 1” in “Meter Status Flag2 byte” to support display two Y axis on chart view • Add connected/disconnect for EVSE tester
4.2.12	03/02/2026	Sylar Lan	• Modified ISO index typo • Add unit index into ISO data
4.2.13	03/26/2026	Sylar Lan	• Update vibration sensor ISO 10816 index
4.2.14	04/14/2026	Sylar Lan	• Modify SV95 function and Unit name ○ function name: OA ○ unit: mm/s RMS or inch/s RMS



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Introduction

This document defined the detail service and reaction needed for METERLiNK 4.x support.

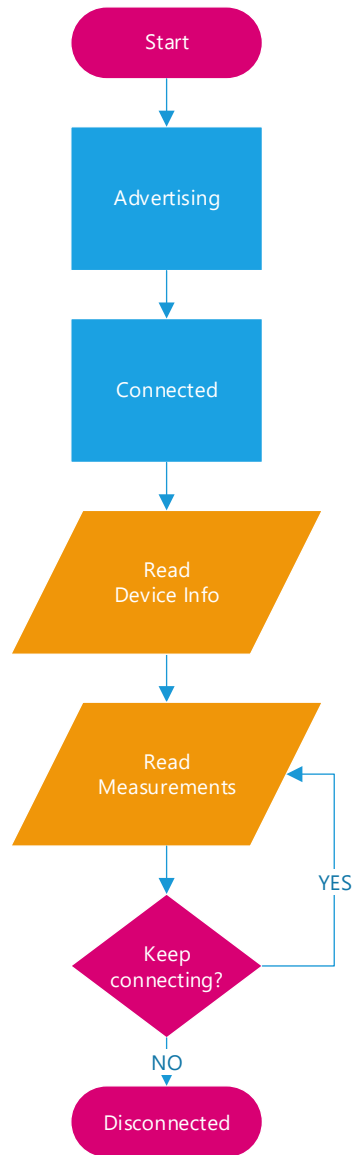
The main project of this document is based on Nordic BLE5 solution nRF52833, the selected module is Raytac , gateway project. The scope of this document will define the advertising, connection, data read/write and security.

System Reference

Currently we are using Nordic Solution for BLE5 connectivity support



Connection Flow





1.0 Advertising

The unit should always deal with advertising with the consideration between discoverable and power consumption.

1.1 Idle (No incoming Connections)

Two kinds of reactions for below HW condition:

1. Host has ability to enable/disable BLE functionality from user interface:
 - a. Always advertising when BLE enabled.
 - b. Stop advertising when connected.
 - c. Start advertising again when disconnected.
2. Host does not have capability to disable BLE functionality from user interface:
 - a. Always advertising when unit had been powered ON.
 - b. Stop advertising when connected.
 - c. Start advertising again when disconnected.

If there is no incoming connection, BLE module should be always on with proper advertising interval with below info:

FLIR brand	EXTECH brand
FLIR <product name> + (4 digi SN)	EXTECH <product name> + (4 digi SN)

Example product name:

- 'FLIR DM93 XXXX', 'FLIR MR77 XXXX', 'FLIR CM78 XXXX', 'FLIR CM83 XXXX'...etc.
- 'EXTECH AN250W XXXX', 'EXTECH LT250W XXXX', 'EXTECH SL250W XXXX', 'EXTECH RH250W XXXX', 'EXTECH RPM250W XXXX'...etc.

Note: XXXX is the last 4 digit number of SN

1.2 With Valid Connection (Pass the Security Rules)

- When APP finds the sensor and starts the connection, the advertising should stop immediately. (and verify the connection with security mechanism. TBD)
- After disconnection, the advertising should be restarted.



2.0 Service definitions

Service definition for generic info and proprietary protocols:

2.1 Generic Services

UUID	Services / Characteristics / Descriptors	
1800	Generic Access	
1801	Generic Attribute	
2A05		Service Changed
2902		Client Characteristic Configuration
180A	Device Information	
2A29		Manufacturer Name String Value: "FLIR System" (fixed)
2A24		Model Number String Value: "EXTECH <product name>"
2A25		Serial Number String - Len: 20 (UTF8s) - Value: "20-<PID>-XXXXXXX"
2A27		Hardware Revision String (Meter Hardware version) Value format: "1.0.0"
2A26		Firmware Revision String (BLE Firmware version) Value format: "0.0.1"
2A28		Software Revision String (Meter Software version) Value format: "0.0.1"
8000		FLIR proprietary, Protocol ID 1 unsigned byte Meter Type 1 unsigned byte



2.2 Generic Services Details

2.2.1 Manufacturer Name String

As the title introduced from generic service definition.

2.2.2 Model Number String

As the title introduced from generic service definition. Fixed with “FLIR Systems”

2.2.3 Serial Number String

As the title introduced from generic service definition.

2.2.4 Hardware Revision String (Meter Hardware version)

As the title introduced from generic service definition. Meter Hardware version revision string.

2.2.5 Firmware Revision String (BLE Firmware version)

As the title introduced from generic service definition. FW revision of BLE module.

2.2.6 Software Revision String (Meter Software version)

As the title introduced from generic service definition. Meter SW revision to indicate protocol version

2.2.7 Protocol ID (Proprietary)

1 unsigned byte which indicates protocol version by digit as below table: (should mapped to ‘SW version’s string)

ML Protocol ID	ML Protocol Version
0x01	METERLiNK v2.0 and prior versions
0x02	METERLiNK v2.1
0x03	METERLiNK v3.0 (Cypress BLE 4.2)
0x04	METERLiNK v4.0 (Nordic BLE 5.0)
0x05	METERLiNK v4.1 (Nordic BLE 5.0 with Bulk data transferring)
0x06	METERLiNK v4.2.0 (Support EVSE serial tester report)
0x07	METERLiNK v4.2.1 (Support Vibration sensor serial)

2.2.8 Meter Type (Proprietary)

1 unsigned byte to indicate meter types, refer to Index table in section [Meter Type Index](#).



2.3 Proprietary Services

UUID	Services(S) / Characteristics(C) / Descriptors (D)
5300 2000 0046 4C49 522D 54414957414E	(S) Sensor Data Service
5300 2002 0046 4C49 522D 54414957414E	(C) Sensor Reading
5300 2010 0046 4C49 522D 54414957414E	(C) METERLiNK RX Characteristic
5300 2011 0046 4C49 522D 54414957414E	(C) METERLiNK TX Characteristic
0000 FE59 0000 1000 8000 00805F9B34FB	(S) DFU Primary Service
8EC9 0001 F315 4F60 9FB8 838830DAEA50	(C) DFU Control Point (Bootloader)
8EC9 0002 F315 4F60 9FB8 838830DAEA50	(C) DFU Packet (Bootloader)
5300 0003 0046 4C49 522D 54414957414E	(C) Buttonless DFU (Application)
5300 0004 0046 4C49 522D 54414957414E	(C) Bonded Buttonless DFU (Application)
METERLiNK v3 Support	
D813 BF66 5E61 188C 3D47 2487320A8B6E	(S) METERLiNK(v3) Service
E9A8 B8C1 B91E 10A1 5241 C4D951378343	(C) METERLiNK Control (Sensor reading)
2902	(D) CCC
00000000-0000-1000-8000-00805F9B34FB	(D) Custom Descriptor

2.3.1 Sensor Reading characteristic

To receive data from meter. For more detail please check “Sensor Reading” chapter

5300 2000 0046 4C49 522D 54414957414E	(S) Sensor Data Service
5300 2002 0046 4C49 522D 54414957414E	(C) Sensor Reading

2.3.2 METERLiNK RX and TX characteristics

To transfer and receive data between BLE and Apps. METERLiNK RX Characteristic: Apps can send commands to BLE via RX characteristics. METERLiNK TX Characteristic: BLE can send commands to Apps via TX characteristic.

5300 2000 0046 4C49 522D 54414957414E	(S) Sensor Data Service
5300 2010 0046 4C49 522D 54414957414E	(C) METERLiNK RX Characteristic
5300 2011 0046 4C49 522D 54414957414E	(C) METERLiNK TX Characteristic

2.3.3 DFU Service

The Primary DFU Service UUID use “FE59” to represent. “FE59” UUID which is registered by Nordic Semiconductor ASA in Bluetooth SIG Assigned Number. We keep this UUID to indicate which BLE chip is used in the meter. We remain default DFU UUID in bootloader mode as well.

0000 FE59 0000 1000 8000 00805F9B34FB	(S) DFU Primary Service
8EC9 0001 F315 4F60 9FB8 838830DAEA50	(C) DFU Control Point (Bootloader)
8EC9 0002 F315 4F60 9FB8 838830DAEA50	(C) DFU Packet (Bootloader)

Meanwhile we change DFU UUID in application layer to prevent our device can be OTA by Noridic OTA tools.

5300 0003 0046 4C49 522D 54414957414E	(C) Buttonless DFU (Application)
5300 0004 0046 4C49 522D 54414957414E	(C) Bonded Buttonless DFU (Application)



3.0 METERLiNK Command

3.1 Device information Characteristic Packet Format

Device information UUID is 180A composed by following data type

Manufacturer Name	Model Name	Serial Number	Meter HW Version
8 bytes	16 bytes	24 bytes	6 bytes

Meter SW Version	BLE FW Version	ML Protocol ID	Mete Type
6 bytes	6 bytes	1 bytes	1 bytes

Manufacturer Name	8 bytes Ascii string to indicate this product is made by "FLIR" or "Extech"
Model Name	16 bytes Ascii string to indicate product name. ex: MR277 or DM286
Serial Number	24 bytes Ascii string to indicate product serial number. - If Meter doesn't have SN. Please fill "0x00", SN will be assigned by BLE - The length of SN number should be longer than 5 Bytes
Meter HW Version	6 bytes Ascii string to indicate meter hardware version. Ex: 1.2.3
Meter SW Version	6 bytes Ascii string to indicate meter software version. Ex: 1.2.3
BLE FW Version	6 bytes Ascii string to indicate BLE Firmware version. Ex: 1.2.3
ML Protocol ID	1 byte, to indicate which version of METERLiNK is supported

ML Protocol ID	ML Protocol Version
0x01	v2.0 and prior versions
0x02	METERLiNK 2.1
0x03	METERLiNK 3.0 (BLE 4.0 Cypress chip)
0x04	METERLiNK 4.0 (BLE 5.0 Nordic chip)
0x05	METERLiNK 4.1 (Support Bulk transferring)
0x06	METERLiNK 4.2(Support EVSE tester report)

Meter Type	1 byte, to indicate Meter type index
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3.2 Sensor Reading Characteristic Packet Format

Sensor Reading information will show on characteristic "5300 **2002** 0046 4C49 522D 54414957414E" characteristic. Total **64 bytes** of reading structure is as below:

Sensor Reading data will keep sending from meter to

Meter Channel	Sensor Channel	Data Description	Meter Status Flag0	Meter Status Flag1	Meter Status Flag2
1 byte	1 byte	16 bytes	1 byte	1 byte	1 byte

Event to APP	Battery Status	Battery Level	Function Name Index	Reading Value	Units Index
1 byte	1 byte	1 byte	1 byte	4 byte	1 byte

Index Number	Sub Units index	Metric Prefix	Digits to Display	Log Date	Maximum OL Vlaue
1 byte	1 byte	1 byte	1 byte	19 bytes	4 bytes

Minimum OL Value	Reserved	CRC
4 bytes	1 bytes	2 bytes

Meter Channel: unsigned, 1 byte

Identifies the number of measurement channels available on the device and ranges from 0x01 to 0xFE.

Example 1:

a meter that reads relative humidity and temperature and reports these readings as exactly two values have two sensor channels.

Example 2:

A meter that reads only temperature, but supports four temperature sensors, has a sensor-channels value of four.

Sensor Channel: unsigned, 1 byte

Identifies the instrument's sensor-reading channel associated with the measurement value contained in the packet. The value ranges from 0 to N-1, where N is the instrument's total number of sensor channels. The number-of-sensor-channels is identified by the Meter Channels (byte 1).

Example 1:

If a meter has only one channel, such as temperature, the temperature reading would be channel 0.

Example 2:

A humidity meter that measures humidity and temperature would have two channels. Channel 0 would be RH and channel 1 would be temperature.



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Data Description, 16 Bytes (Big Endian)~~(Not supported in METERLiNK APP, but Meter has to support it)~~

- Please check **Data Description bit** in **Meter Status Flag2** byte to decide this filed function.
 - “**Data Descrtipon bit = 0**” in **Meter Status Flag2** byte
 - Data Description won’t apply below structure, Meter can fill and information to this filed with ASCII code, in default case all of byte should be 0x00.
 - “**Data Descrtipon bit = 1**” in **Meter Status Flag2** byte
 - To enable below data structure field for some special case

For the special project with sensor data

- **EVSE Tester Project**

1. **Enable Data Descrtipon bit = 1" in Meter Status Flag2 byte**
2. Byte 15 to set 0x01 to indicate App should read following ASCII byte data
3. EVSE Tester should show "A" to "F" ascii into this filed
 - A: Not Connected, show **Not connected icon**
 - B: Connected, Not Charging, **depends on EV86 state**
 - C: Charging Ready, Ventilation not required, **depends on EV86 state**
 - D: Charging Ready, Ventilation required, **depends on EV86 state**
 - E: Error, Disconnected or Fault, **Not connected icon**
 - F: Error from EVSE, show **Not connected icon**



Not connected



Connected

Byte	Byte 15	Byte 14	Byte 13	Byte 12	Byte 11	Byte 10	Byte 9	Byte 8	Byte 7	Byte 6	Byte 5	Byte 4	Byte 3	Byte 2	Byte 1	Byte 0
None	0x00	Default 0x00														
Ascii	0x01	Ascii to descript data (Big Endian)														

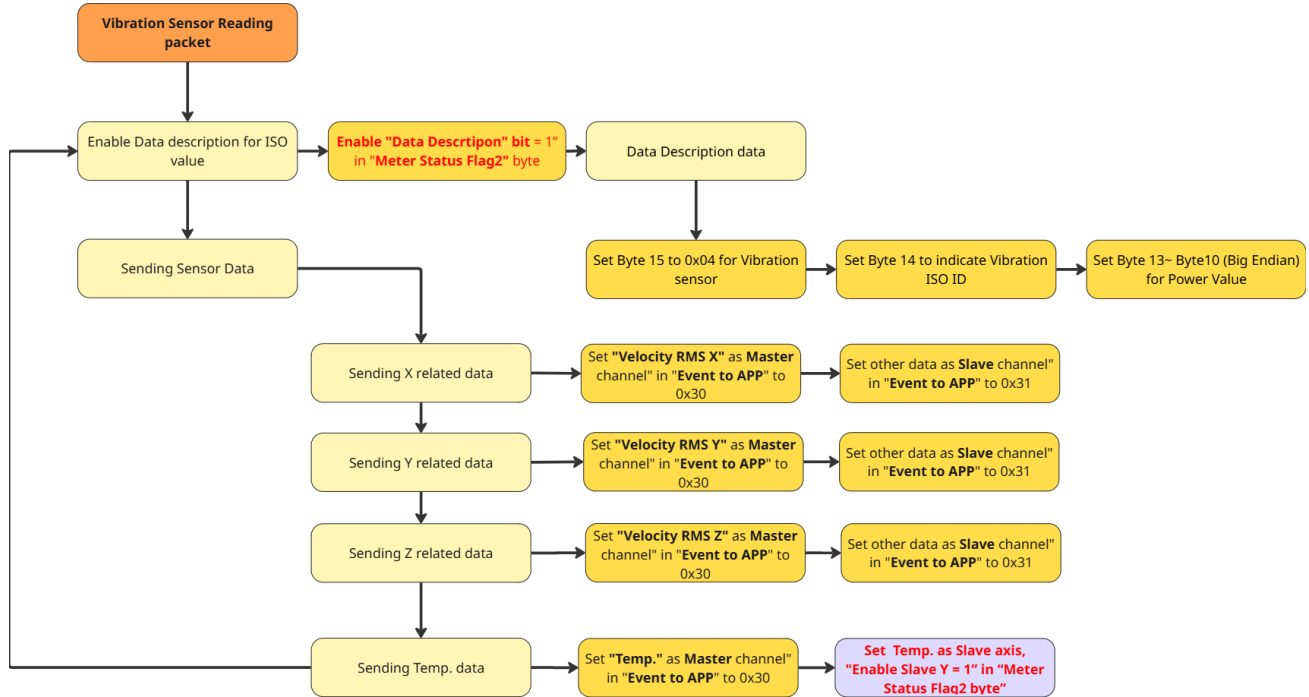
4. Function name index: 0x97
5. Fill 0x00 into the items below
 - a. Reading Value
 - b. Unit index
 - c. Index number
 - d. Sub unit index
 - e. Metric Prefix
 - f. Digits to Display
 - g. Maximum OL Vlaue
 - h. Minimum OL Value

- **Vibration sensor Project**

1. **"Enable Data Descrtipon bit = 1" in "Meter Status Flag2" byte**
2. Byte 15 to set 0x04 to indicate App this is Vibration sensor
3. Byte 14 will indicate vibration ISO ID
4. Byte 13 ~ Byte 10 (Big Endian) are for Power, using Value Float32 (IEEE-754-2008) for indication
5. Log Data default date is **1970:01:01, 00:00:00**
6. **Set Temp. data as Slave Y (right side of trend chart) by "Enable Slave Y = 1" in "Meter Status Flag2 byte"**



- Vibration sensor reading packet



- Vibration sensor reading function and unit index

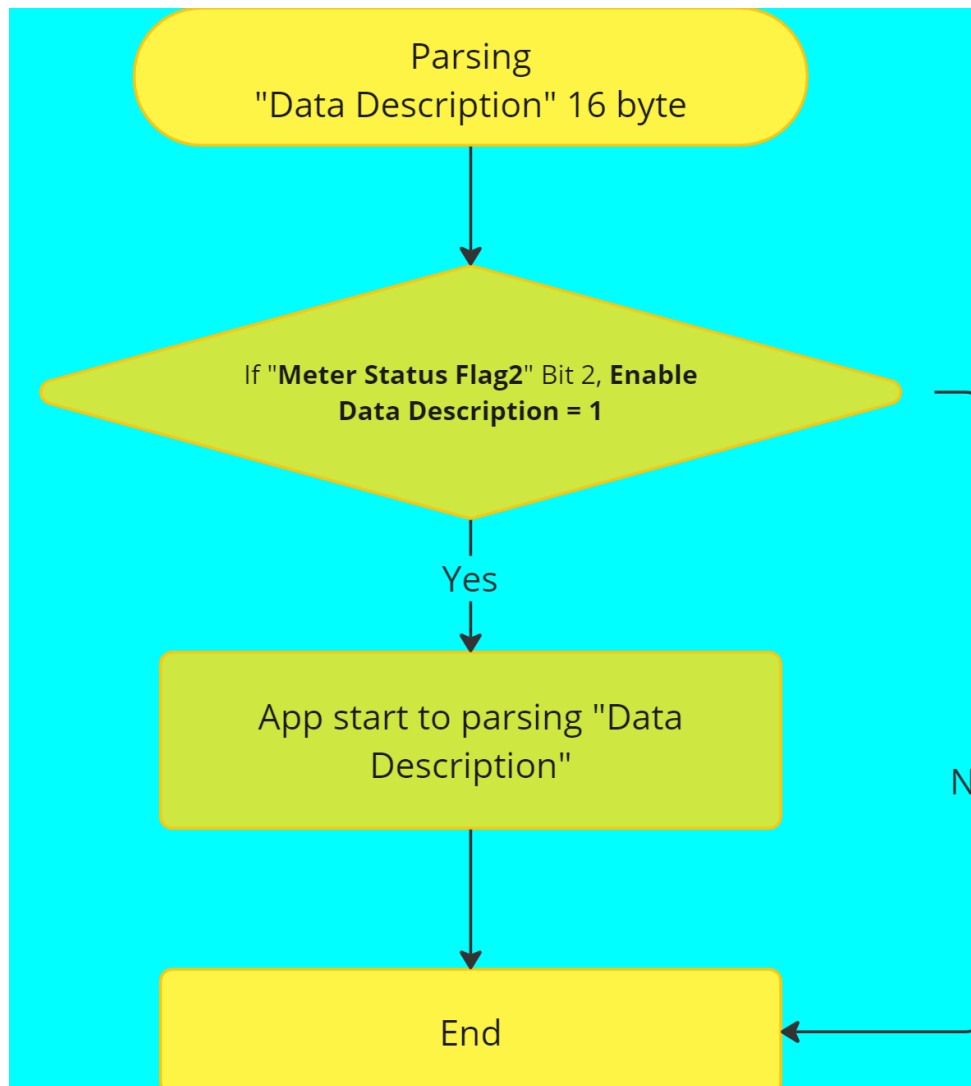
	Name	Unit	Function index	Function Name	Unit index	Unit Name
Temp.	Temp.	°C	0x43	Temperature	0x14	°C
		°F	0x43	Temperature	0x15	°F

	Name	Unit	Function index	Function Name	Unit index	Unit Name
X	Velocity RMS	mm/s	0xA1	Velocity RMS -X	0x56	mm/s
	Acceleration RMS	g	0x9D	Accel RMS	0x55	g
	Acceleration Mean	g	0x9E	Mean	0x55	g
	Acceleration Median	g	0x9F	Median	0x55	g
	Acceleration (0 to Peak)	g	0xA0	Acceleration (0 to Peak)	0x55	g
	Displacement (Peak to Peak)	um	0x99	Displacement (Peak to Peak)	0x2d	um
	Kurtosis	n/a	0x9A	Kurtosis	0x00	unit-less
	Crest Factor	n/a	0x2F	Crest factor	0x00	unit-less
	Skewness	n/a	0x9B	Skewness	0x00	unit-less
	Deviation	n/a	0x9C	Deviation	0x00	unit-less



Y	Name	Unit	Function index	Function Name	Unit index	Unit Name
	Velocity RMS	mm/s	0XA2	Velocity RMS -Y	0x56	mm/s
	Acceleration RMS	g	0X9D	Acceleration RMS	0x55	g
	Acceleration Mean	g	0X9E	Mean	0x55	g
	Acceleration Median	g	0x9F	Median	0x55	g
	Acceleration (0 to Peak)	g	0xA0	Acceleration (0 to Peak)	0x55	g
	Displacement (Peak to Peak)	um	0x99	Displacement (Peak to Peak)	0x2d	um
	Kurtosis	n/a	0x9A	Kurtosis	0x00	unit-less
	Crest Factor	n/a	0x2F	Crest factor	0x00	unit-less
	Skewness	n/a	0x9B	Skewness	0x00	unit-less
	Deviation	n/a	0x9C	Deviation	0x00	unit-less

Z	Name	Unit	Function index	Function Name	Unit index	Unit Name
	Velocity RMS	mm/s	0XA3	Velocity RMS - Z	0x56	mm/s
	Acceleration RMS	g	0X9D	Acceleration RMS	0x55	g
	Acceleration Mean	g	0X9E	Mean	0x55	g
	Acceleration Median	g	0x9F	Median	0x55	g
	Acceleration (0 to Peak)	g	0xA0	Acceleration - (0 to Peak)	0x55	g
	Displacement (Peak to Peak)	um	0x99	Displacement (Peak to Peak)	0x2d	um
	Kurtosis	n/a	0x9A	Kurtosis	0x00	unit-less
	Crest Factor	n/a	0x2F	Crest factor	0x00	unit-less
	Skewness	n/a	0x9B	Skewness	0x00	unit-less
	Deviation	n/a	0x9C	Deviation	0x00	unit-less





Function	Byte	Byte		Byte	Byte	Byte	Byte	Byte	Byte	Byte	Byte	Byte	Byte	Byte	Byte	Byte	
	15	14		13	12	11	10	9	8	7	6	5	4	3	2	1	0
None	0x00	Default 0x00															
Ascii	0x01	Ascii to descript data (Big Endian)															
N/A	Type Index	Functions selection		Show Fun. String	Show Unit Stting	Float 32 (4 bytes) data				Show Status	Show Info	Reserved					
Watt OL	0x02	0x00	Null	Show Fun. String	Show Unit Stting	Float 32 (4 bytes) data				N/A	N/A	N/A					
		0x01	Max OL														
		0x02	Min OL														
Insulation	0x03	0x00	Null	0x00 = No show	(0x53) Ratio	Float 32 (4 bytes) data If “show Unit index” == “0x00” Data filed will be “0x00” equal to no data				0x00 = No show	N/A	N/A					
		0x01	Manual							0x01 = Pass,							
		0x02	PI time rate: 10m/1m							0x02 = Fail,							
		0x03	DAR time rate: 1m/30s							0x03 = Error.							
		0x04	DAR TR: 1m/15s														Check COMP Table
		0x05	COMP	0x00 = No show													
Vibration sensor																	
Function	Byte	Byte		Byte	Byte	Byte	Byte	Byte	Byte	Byte	Byte	Byte	Byte	Byte	Byte	Byte	Byte
	15	14		13	12	11	10	9	8	7	6	5	4	3	2	1	0
Vibration	0x04	ISO Index		Value Float32 (IEEE-754-2008)				Unit index									
EVSE	0x05	Connected =0x01 Disconnect = 0x02		Ascii to descript data (Big Endian)													

• Type Index (Byte 15)

- Index = 0x01, Meter will send ASCII code to APP, then app should show that message on App if necessary.
- Index = 0x02, to support Watt OL special case (TBD)
for example, "Watt" have 4 type of OL value as following table, but the "Function" and "Unit" is different to Watt

Function	Fun. index	Unit Index
OL DC-V	DC	V
OL DC-A	DC	A
OL AC-V	AC	V
OL AC-A	AC	A



- Index = 0x03, to support PI, DAR and COMP function
 - Regarding PI, DAR, COMP, the original unit is Ohm. But the testing result will be including
 1. Pass/Fail or Error
 2. Ratio value
 3. COMP

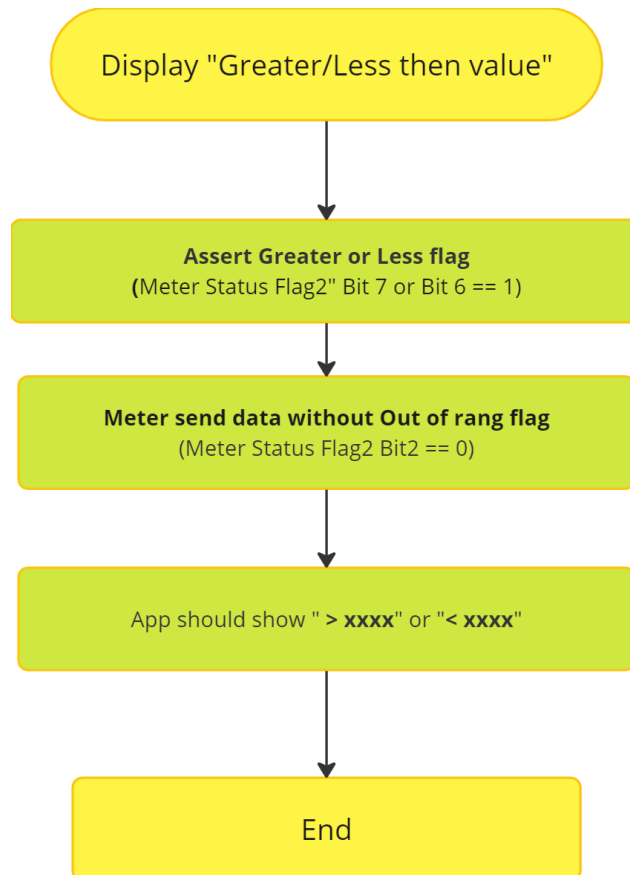
To show extra information's on METERLiNK App and indicate which function is used. Meter should fill correct data with the protocol definition.

(Byte 13) Show Fun. String	Show function string in extra info field. If this byte == 0x00, App won't show this string. If this value is not 0x00. App should show corresponding index string
(Byte 12) Show Unit String	Show Unit index string in extra info field. If this byte == 0x00, App won't show this string. If this value is not 0x00. App should show corresponding index string
(Byte 11 to Byte 8) Float32	Show float data on app. If "show unit string" is 0x00, this value won't show on App
(Byte 7) Show Status	Show Status string in extra info field. If this byte == 0x00, App won't show this string. If this value is not 0x00. App should show corresponding index string
(Byte 6) Show Info	Show Info. string in extra info field. If this byte == 0x00, App won't show this string. If this value is not 0x00. App should show corresponding index string

- Byte 6, Show info
 - Compare status string table

Index	COMP. Display table
0x01	COMP, 100K (Ohm)
0x02	COMP, 200K (Ohm)
0x03	COMP, 500K (Ohm)
0x04	COMP, 1M (Ohm)
0x05	COMP, 2M (Ohm)
0x06	COMP, 5M (Ohm)
0x07	COMP, 10M (Ohm)
0x08	COMP, 20M (Ohm)
0x09	COMP, 50M (Ohm)
0x0a	COMP, 100M (Ohm)
0x0b	COMP, 200M (Ohm)
0x0c	COMP, 500M (Ohm)

- To support this information, Meter developer should use index 0x03 to give more information to App.
 - In PI, DAR, COMP function, App won't display OL mark instead of below
 - "> value"
 - "< value"





Meter Status: unsigned, 4 bytes

- **Meter Status Flag0** – Various indicators on the meter.

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Reserved	Error	Alarm	Peak	Max	Min	Hold	Negative

bit 7 **Reserved:**

bit 6 **Error:** 1 = error flag raised, 0 = no errors.

Note: Not support test lead wrong insertion

bit 5 **Alarm:** 1 = alarm flag raised, 0 = no alarm.

bit 4 ***Peak:** To detect AC capture response time is 100ms

1 = peak value saved, 0 = peak value not saved.

Peak	Max	Min	Note
1	1	0	• The sensor reading value will be Peak Max value. • Flag should assert once only • App should handle this flag (TBD for METERLiNK App) • Once Peak Max sensor reading packet is sent, Meter should send “Real” Sensor Reading value immediately.
1	0	1	• The sensor reading value will be Peak Min value. • Flag should assert once only • App should handle this flag (TBD for METERLiNK App) • Once Peak Min sensor reading packet is sent, Meter should send “Real” Sensor Reading value immediately.
1	0	0	(Status) • The sensor reading value will be “Real” sensor reading value. • Peak flag will keep asserted until user disable it. • App should handle this flag (TBD for METERLiNK App) • In Peak mode, it just a status to indicate App Meter in Peak mode

bit 3 **Max:** 1 = maximum value saved, 0 = maximum value not saved.

- While “Max” is arrested, the Sensor Reading value is the Max value
- **Flag should assert once only**
- App should handle this packet separately (TBD for METERLiNK App)
- Once Max sensor reading packet is sent, Meter should keep sending Sensor Reading value immediately.

bit 2 **Min:** 1 = minimum value saved, 0 = minimum value not saved.

- While “Min” is arrested, the Sensor Reading value is the Min value



- **Flag should assert once only**
- App should handle this packet separately (TBD for METERLiNK App)
- Once Min sensor reading packet is sent, Meter should keep sending Sensor Reading value immediately.

bit 1 **Hold:**

1 = meter is on hold. LCD values will not change.
0 = meter is not on hold. LCD values are dynamic

- While “Hold” is arrested, the Sensor Reading value is the Hold Value
- **Flag should assert once only**
- App should handle this packet separately (TBD for METERLiNK App)
- Once Hold sensor reading packet is sent, Meter should keep sending “Real Sensor Reading” value immediately.

bit 0 **Negative:** Indicates the measurement value is negative; non-controlling, used as a ‘helper’ indication when displaying the ASCII measurement string (i.e. if value is negative, the bit is set; if the value is positive, the bit is cleared).

1 = flag on, reading value is a negative
0 = flag off, reading value is a positive

▪ **Meter Status Flag1** - Various indicators on the meter.

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Warning	Peak-to-Peak	RMS	Average	Difference	Out-of-Range	Active/Blink	Relative

bit 7 **Warning:** default value = 0

bit 6 **Peak-to-Peak:** 1 = indicates a peak-to-peak measurement value

bit 5 **RMS:** 1 = indicates a RMS measurement value (AC only)

- While “RMS” is arrested, the Sensor Reading value is the RMS value

bit 4 **Average:** 1 = indicates a average measurement value. (AC only)

- While “Average” is arrested, the Sensor Reading value is the Average value

bit 3 **Difference:** 1 = indicates the difference between two measurement values. Will collate with Relative (bit 0)

bit 2 **Out-of-Range:** 1 = indicates measurement value is out-of-range

- While Out-of-Range is asserted. Meter should keep sending “Real Sensor Reading” value
- Please fill correct “digits to display” and “Metric Prefix” as well

bit 1 **Active/Blink:** 1 = Blink, 0 = Active

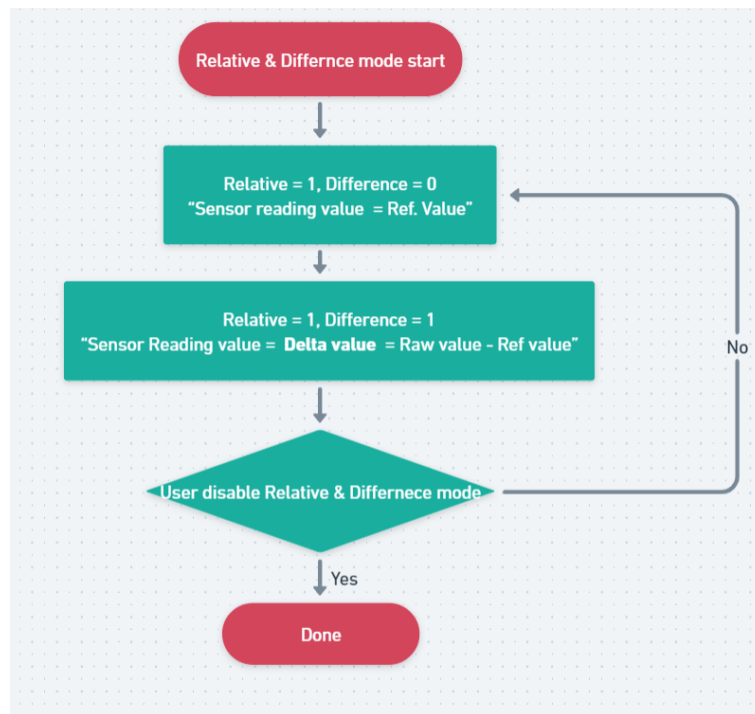
bit 0 ***Relative:** 1 = Relative is on, 0 = Relative is off

While **Relative** is arrested, there have 2 situations in Relative mode, as following shown



Relative	Difference	Note
1	0	The sensor reading value will be relative value. sensor reading value = Ref. value
1	1	The Sensor Reading value as below formula. Sensor Reading value = Delta, value = Raw value - Ref value

- Meter will follow the process below to send Relative & difference value to App



▪ Meter Status Flag2

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Greater then value	Less the value	Reserved	Reserved	Enable Slave Y	Enable Data Description	Fast Sampling Rate	Slow Sampling Rate

bit 7 **Greater than value:** default value = 0. Greater than value = 1

bit 6 **Less the value:** default value = 0. Less the value = 1

bit 5 **Reserved:** default value = 0

bit 4 **Reserved:** default value = 0



bit 3 Enable Slave Y:

default value = 0

0 = The trend chart show main Y axis at left side

1 = Enable slave Y axis, the trend chart will show main Y axis at left side, slave Y axis at right side on the trend chart view

bit 2 Enable Data Description:

1 = Enable Data Description 16byte data

0 = Disable Data Description 16byte data

bit 1 Fast Sampling Rate: default value = 0

1 = indicates value is measured by fast sampling rate

bit 0 Slow Sampling Rate: default value = 0

1 = indicates value is measured by slow sampling rate

▪ **Event to APP**

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Event to App							

“Event to App” byte

0x00	No event
0x01	“Stop Sensor Reading” and stop recording data, for some of device which is not powerful. Sensor Reading data may not be sent in some of situation simultaneously. - Case1: MCU is busy and without resources to handle others’ action. In this situation meter can sent this command to App to indicate “Sensor Reading” data is going to stop. App will also stop recording and drawing charts and indicate that user Sensor reading is stopped. - Case2: While CM57 is going to transfer file to app. Meter can send this command to indicate "Sensor Reading" will stop transferring. - Once “Stop Sensor Reading” process is done. Meter should set “Event to app” byte to 0x00 (No event). And meter should send “Real Sensor Reading” packet immediately.
0x02	Last reading data, App should stop drawing and keep current chart
...	
0x30	Grouping Master
0x31	Grouping Slave
...	
0xFF	End of Event to App



Battery Status: unsigned 1 byte

Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Auto Power Off	Reserved	Reserved	Reserved	Simple battery level	Battery Temp. warning	Charging	Low battery

bit 7 **Auto Power Off:** 1 = On, 0 = Off

bit 6 **Reserved:**

bit 5 **Reserved:**

bit 4 **Reserved:**

bit 3 **Simple Battery Level:** 1 = indicates only support simple battery level

bit 2 **Battery Temp. Warning:** 1 = indicates temperature is out of range

bit 1 **Charging:** 1 = Device is charging

bit 0 **Low Battery:** 1 = indicates battery is in low battery, low battery percentage will depend on different product specifications

Battery Level: unsigned 1 byte

If **Simple Battery Level is not set**, Battery Level will use 0 ~ 100 percentage to represent battery level. Once **Simple Battery Level is set to 1**, Battery level will follow below table to represent simple battery level.

Bit7 ~ Bit4	Bit3 ~ Bit0
Maximum Simple battery level	Simple Battery level

bit7-Bit4 **Maximum Simple battery level:** indicate the maximum simple battery level which can be shown. (maximum value = 5)

bit3-Bit0 **Simple Battery Level:** indicate the current simple battery level. The maximum display level is dependent on "Max Simple battery level "

Function Name Index: unsigned, 1 byte

Index into the Function Name Index table identifying the sensor-reading channel name. Index values start at zero. See [Function Name Index](#) table at end of document.

Reading Value: float32, 4 Bytes

Meter reading represented as a float value IEEE-754-2008. The unit will follow **Metric Prefix** (Ref: <http://www.h-schmidt.net/FloatConverter/IEEE754.html>)



Units Index: unsigned, 1 byte

Index into the Measurement Units Index table identifying the sensor-reading measurement units. Index values start at zero. See Measurement Units Index table at end of document.

Index Number: unsigned, 1 byte

For instruments with multiple channels of similar functionality (i.e. same Function Name Index values), the Index Number provides a mechanism to distinguish between these channels. The Index number is used to uniquely reference each channel. Index values start at zero.

Example:

A 5-channel temperature instrument will contain the same Function Name Index value (e.g. 0x43 for temperature) for each channel. To help distinguish between these readings, the Index Number is used. In this example, the Index Number will range from 0 to 4

Subunits index: unsigned, 1 byte

Subunit category for “Units Index” (Byte 29) to indicate same group of units. Default value is 0x00 to indicate the main unit’s name.

Example:

Units index	Unit name	Subunits index	Subunits Name
0x11	dB		
		0x00 (Default)	dB
		0x01	dBm
		0x02	dB(A)
		0x03	dB(B)
		0x04	dB(C)

Metric Prefix: signed, 1 byte

Unit prefix (m, u, n, k, M, G, T) used to indicate how the reading is displayed. Note, this is a non-controlling indicator for display purposes only.

Valid entries: -9 = n, -6 = u, -3 = m, 0 = none, 3 = k, 6 = M, 9 = G, 12 = T

Example:

floating point value of 1234.5678 with a prefix of 3, indicates the reading value should be displays as “1.23 k

Digits to Display: unsigned, 1 byte

Identifies the number of digits used to display the sensor-reading value on the instruments display. A leading zero used before a decimal point is not included in the count.



Examples:

floating pt value 123.45678	3 digits to display: 123 4 digits to display: 123.4
floating pt value 0.123456	3 digits to display: 0.123 4 digits to display: 0.1234
floating pt value 0.012000	3 digits to display: 0.012 4 digits to display: 0.0120
floating pt value 0.000000	3 digits to display: 0.000 4 digits to display: 0.0000
value 0	0 digits to display: 0
value 0.0	1 digits to display: 0.0
value 1	0 digit to display: 0
value 1	1 digit to display: 1
value 6000	4 digit to display: 6000
Note: if the digits display = 0, It will only show 0 on app.	

- In METERLiNK app should handle rounding of the number

Example:

value = 0.22999398	2 digit to display: 0.23
value = 0.22199398	2 digit to display: 0.22

Log Date: ASCII string, 19 Bytes

Used when reading the date and time of logged information. The date format used is in similar accordance with ISO (8601) date and time format, which is YYYY-MM-DD and HH:MM: SS. The ISO date format is universally understandable. Note that time zones are not specified. If not getting logged data, 40-58 bytes should always be 0x00.

Example (date):

2013-07-13 would equate to.
byte 40-43 YYYY = 0x32 0x30 0x31 0x33
byte 44 Hyphen = 0x2D
byte 45-46 MM = 0x30 0x37
byte 47 Hyphen = 0x2D
byte 48-49 DD = 0x31 0x33
Byte 50 Vertical Bar = 0x7C

Example (timestamp):

23:23:02 would equate to
byte 51-52 HH = 0x32 0x33
byte 53 Colon = 0x3A
byte 54-55 MM = 0x32 0x33



byte 56 Colon = 0x3A
byte 57-58 SS = 0x30 0x32

Maximum OL Value: float32, 4 Bytes

To indicate Meter maximum OL reading value, represented as a float value IEEE-754-2008. The unit will follow **Metric Prefix** (Ref: <http://www.h-schmidt.net/FloatConverter/IEEE754.html>)

App should replace this value with corresponding default maximum OL value.)

Minimum OL Value: float32, 4 Bytes

To indicate Meter minimum OL reading value, represented as a float value IEEE-754-2008. The unit will follow **Metric Prefix** (Ref: <http://www.h-schmidt.net/FloatConverter/IEEE754.html>)

(Note: EXTECH SL250W, LT250W, RH250W, RPM250W, AN250W products don't support this field. App should replace this value with corresponding default minimum OL value.)

Reserved: TBD, 1 Bytes

CRC: unsigned byte, 2 Bytes, **Big-endian**,

CRC calculation of the 62-bytes of sensor reading data using the CRC-16/CCITT-FALSE algorithm based on the polynomial $X^{16}+X^{12}+X^5+1$ A value of zero in this location will disable the CRC check

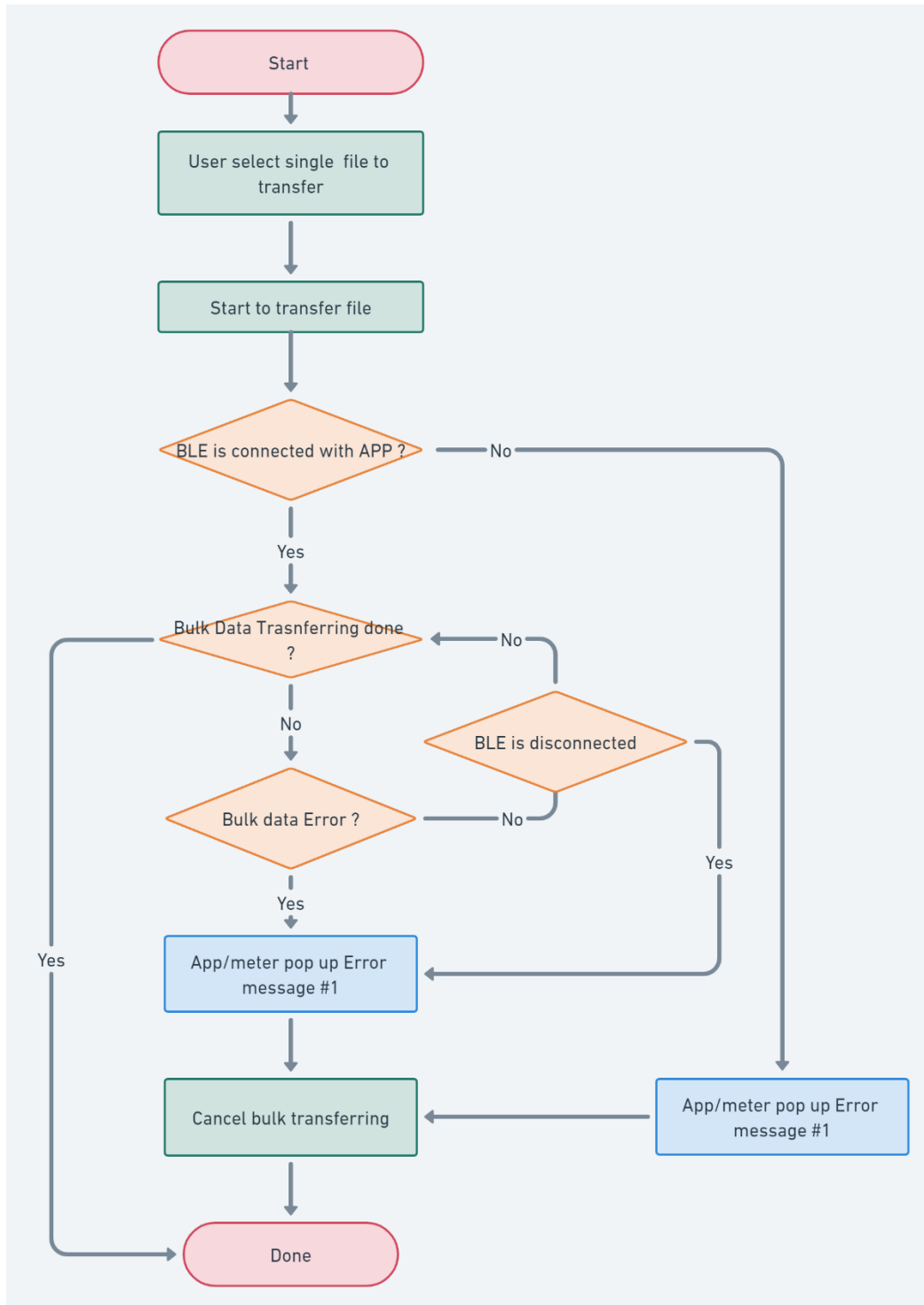
Poly: 0X1021
Init: 0XFFFF
Xorout: 0X0000

Ref. <https://crccalc.com/>

3.3 Bulk Data Transferring

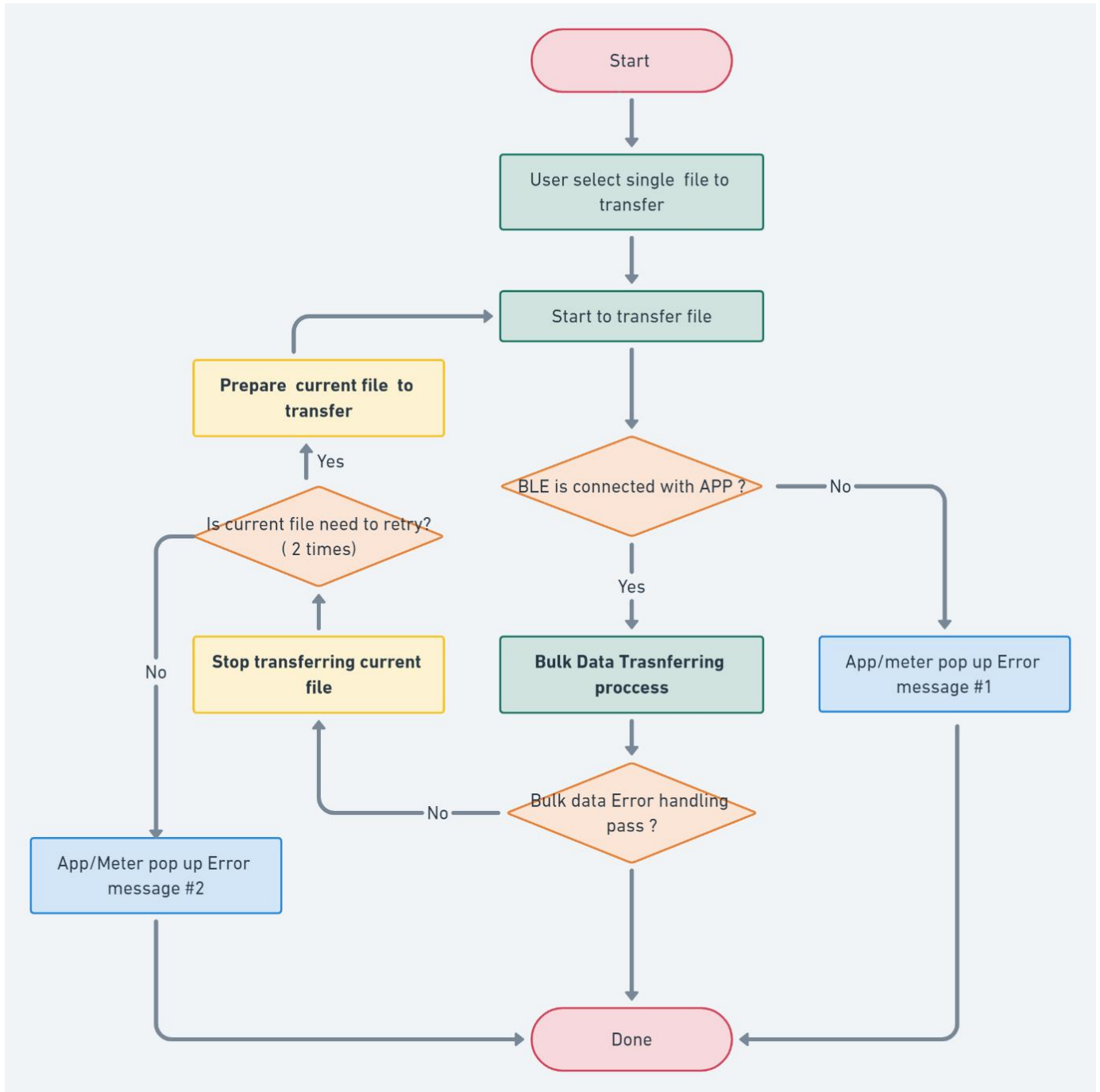
The main purpose of Bulk data transferring is to transfer data from meter to METERLiNK Application, the file format including image, log, video and any others file format. Meter should follow the communication transferring process to send files to METERLiNK app.

3.3.1 Bulk Data Single File Transferring Flow Overview (without retry handling)



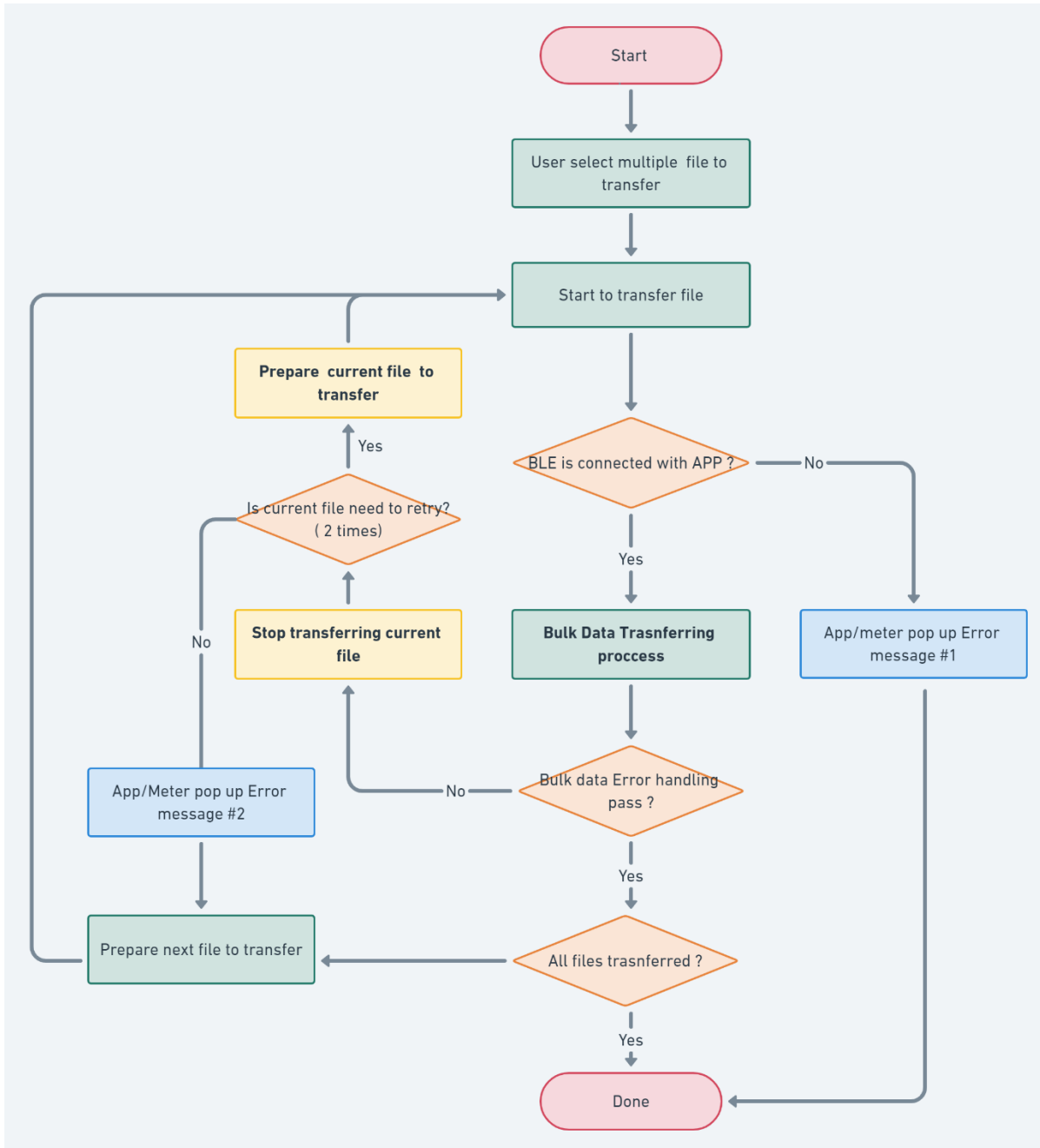
Note: Regarding “Bulk Data Transferring process” detail, please check chapter 3.3.4

3.3.2 Bulk Data Single File Transferring process with retry handling (Not support Now)



Note: Regarding “Bulk Data Transferring process” detail, [please check chapter 3.3.4](#)

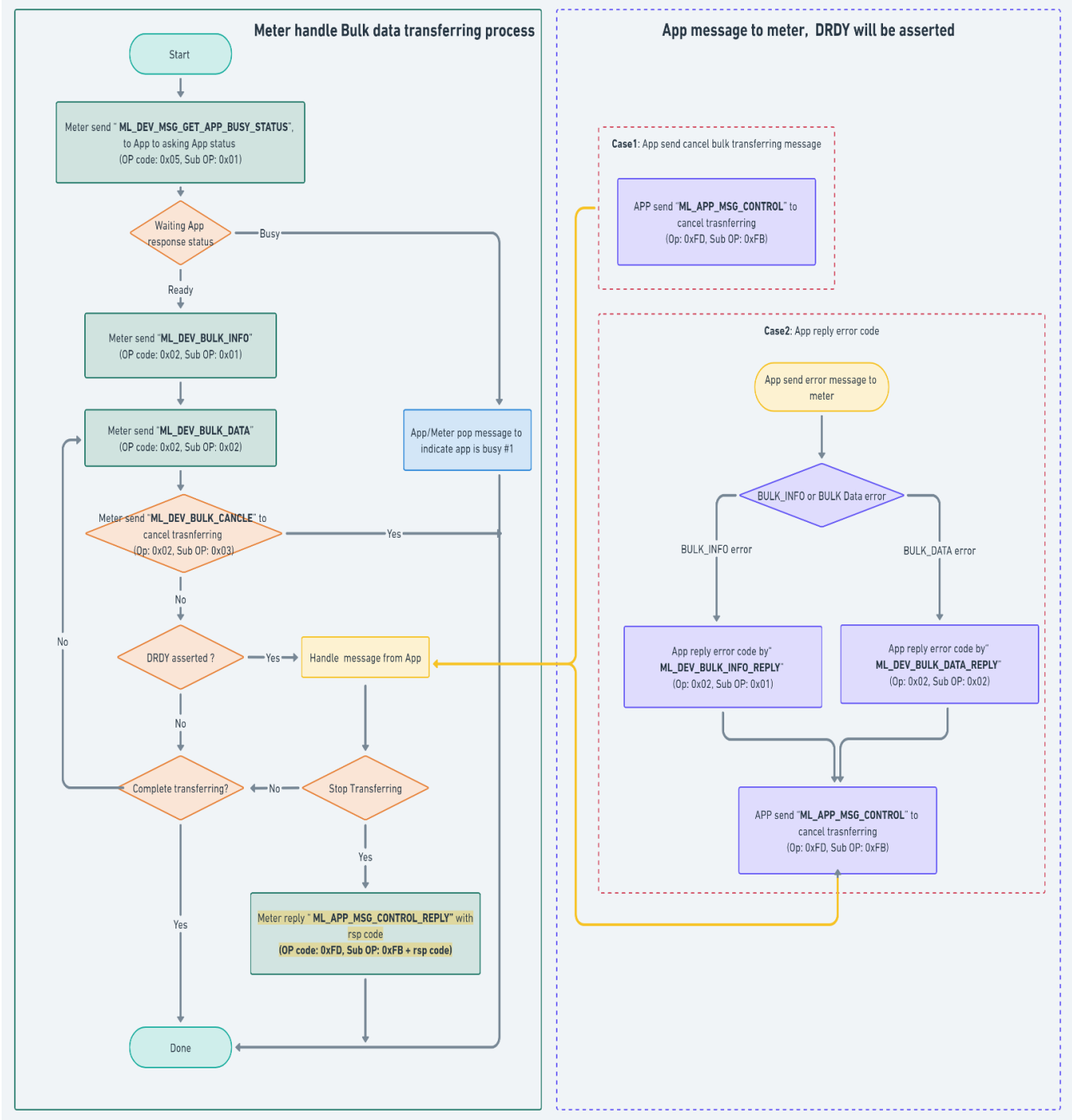
3.3.3 Bulk Data Multiple File Transferring process retry handling (Not support Now)



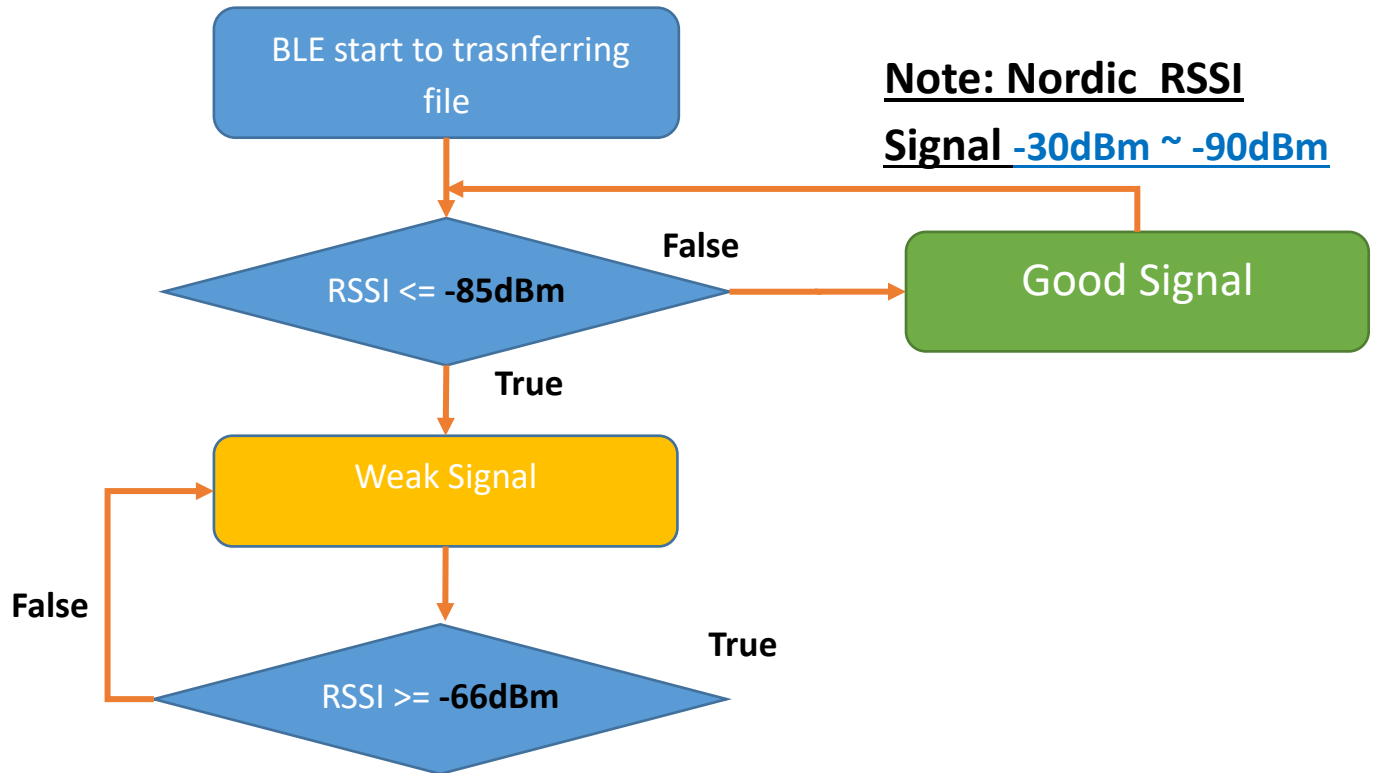
Note: Regarding “Bulk Data Transferring process” detail, please check chapter 3.3.4



3.3.4 Bulk Data File Transferring Process Detail



3.3.4 RSSI Weak Signal Notification



BLE RSSI Weak Signal Control



3.4 Metrolink Tx Commands

BLE will send data via “Metrolink TX Characteristic” Metrolink App. In other words, METERLiNK App should listen notification from “Metrolink TX Characteristic”.

5300 2011 0046 4C49 522D 54414957414E	(C) METERLiNK TX Characteristic
--	---------------------------------

3.4.1 Engineer Packet lost test Command

Command from meter to app to verify **packet error** and **packet sequential error**. All the **endianness** is **Big-endian**

- **Packet Error:**
 - In one packet there have byte lost or byte incorrect. It will be checked by CRC. While CRC errors happened, it will return CRC error code by SPI command. BLE won't send this packet to App. And waiting for Meter to send this packet again.
- **Packet Sequential Error**
 - For example, Meter sends 10 packets to App. Each packet has its sequential number. In some case the 10 of the packets maybe lost 1 packet. We call it packet sequential error.

- **ML_DEV_ENG_PACKEET_LOST_TEST**

Command from meter to App. to verify **packet error** and **packet sequential error**

OP	Sub OP	Data Length	Pad	Data	CRC
1 byte	1 byte	1 byte	1 byte	Depend on Data Lenght	2 bytes

OP 0x00

Sub OP 0x01 Verify bulk data packet lost rate

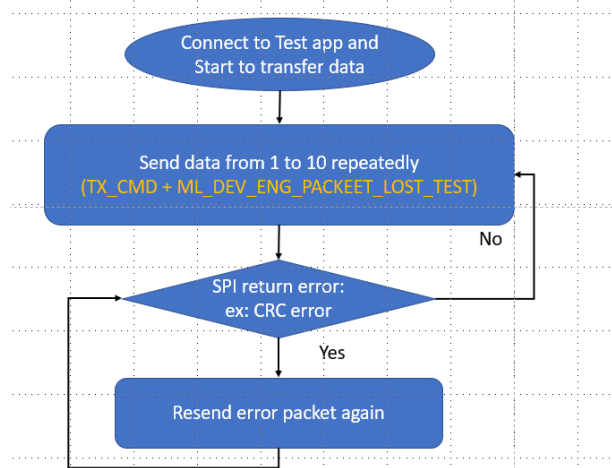
Data Length Data length: Will depend on Maximum MTU size,
(Data Length = **MTU Size** - OP (1 byte) - Sub OP (1 byte) - Data Length (1 Byte) - Pad (1 byte) - CRC (2 byte))
for more detail, please check “BLE METERLiNK Host SPI-bus Commands 1.XX.pdf”.
Note: BLE Maximum size is 244 bytes

Pad 0x00, Memory aligned

Data Meter should send data from "0x01" to "0x0A" repeatedly. And App will calculate packet loss rate and sequence lost rate

CRC Using the CRC-16/CCITT-FALSE, calculating data from OP code field to Data

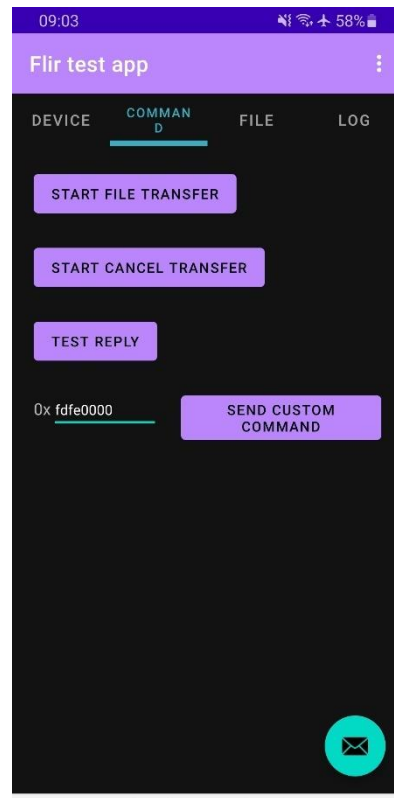
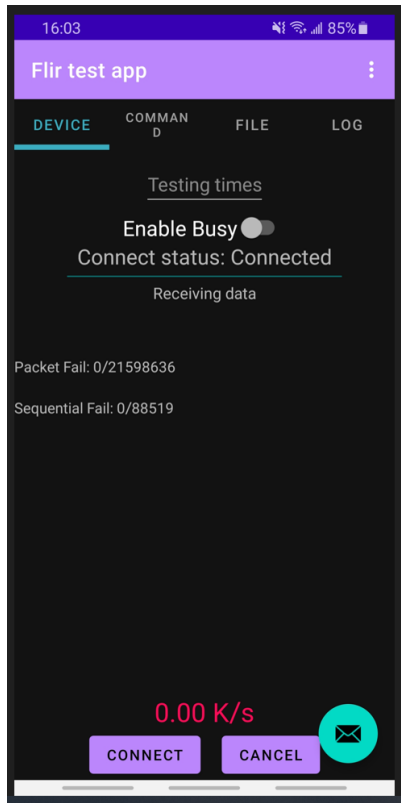
Test process and command detail



- App for verify ML_DEV_ENG_PACKEET_LOST_TEST command



- Test App APK on Android (Please ask FLIR to share this APK for driver developer). This app can detect Packet CRC errors and packet sequential errors. It also shows the throughput on app for Meter developers to tuning throughput.
- In Command page, developers can send App command to Meter to verify DRDY function and the data which comes from App is correct.



- Example SPI Command and Log

TX cmd (8 Byte) + Bulk Data (244 Byte) =

[2023-02-03 14:18:46.725]	DEBUG:	send bulk cmd 0x1,0x40,0xf4,0x0,0x0,0x0,0x0,0x4,0x0,0x1,0xee,0x0,0x1,0x1,0x1,0x1 0x01
[2023-02-03 14:18:46.739]	DEBUG:	send bulk cmd 0x1,0x40,0xf4,0x0,0x0,0x0,0x0,0x4,0x0,0x1,0xee,0x0,0x2,0x2,0x2,0x2 0x02
[2023-02-03 14:18:46.754]	DEBUG:	send bulk cmd 0x1,0x40,0xf4,0x0,0x0,0x0,0x0,0x4,0x0,0x1,0xee,0x0,0x3,0x3,0x3,0x3 0x03
[2023-02-03 14:18:46.768]	DEBUG:	send bulk cmd 0x1,0x40,0xf4,0x0,0x0,0x0,0x0,0x4,0x0,0x1,0xee,0x0,0x4,0x4,0x4,0x4 0x04
[2023-02-03 14:18:46.784]	DEBUG:	send bulk cmd 0x1,0x40,0xf4,0x0,0x0,0x0,0x0,0x4,0x0,0x1,0xee,0x0,0x5,0x5,0x5,0x5 0x05
[2023-02-03 14:18:46.798]	DEBUG:	send bulk cmd 0x1,0x40,0xf4,0x0,0x0,0x0,0x0,0x4,0x0,0x1,0xee,0x0,0x6,0x6,0x6,0x6 0x06
[2023-02-03 14:18:46.813]	DEBUG:	send bulk cmd 0x1,0x40,0xf4,0x0,0x0,0x0,0x0,0x4,0x0,0x1,0xee,0x0,0x7,0x7,0x7,0x7 0x07
[2023-02-03 14:18:46.828]	DEBUG:	send bulk cmd 0x1,0x40,0xf4,0x0,0x0,0x0,0x0,0x4,0x0,0x1,0xee,0x0,0x8,0x8,0x8,0x8 0x08
[2023-02-03 14:18:46.843]	DEBUG:	send bulk cmd 0x1,0x40,0xf4,0x0,0x0,0x0,0x0,0x4,0x0,0x1,0xee,0x0,0x9,0x9,0x9,0x9 0x09
[2023-02-03 14:18:46.858]	DEBUG:	send bulk cmd 0x1,0x40,0xf4,0x0,0x0,0x0,0x0,0x4,0x0,0x1,0xee,0x0,0xa,0xa,0xa,0xa 0x0a
[2023-02-03 14:18:46.858]	DEBUG:	send bulk cmd 0x1,0x40,0xf4,0x0,0x0,0x0,0x0,0x4,0x0,0x1,0xee,0x0,0xa,0xa,0xa,0xa
[2023-02-03 14:18:46.873]	DEBUG:	send bulk cmd 0x1,0x40,0xf4,0x0,0x0,0x0,0x0,0x4,0x0,0x1,0xee,0x0,0x1,0x1,0x1,0x1
[2023-02-03 14:18:46.889]	DEBUG:	send bulk cmd 0x1,0x40,0xf4,0x0,0x0,0x0,0x0,0x4,0x0,0x1,0xee,0x0,0x2,0x2,0x2,0x2
[2023-02-03 14:18:47.54]	DEBUG:	send bulk cmd 0x1,0x40,0xf4,0x0,0x0,0x0,0x0,0x4,0x0,0x1,0xee,0x0,0x3,0x3,0x3,0x3
[2023-02-03 14:18:47.64]	DEBUG:	send bulk cmd 0x1,0x40,0xf4,0x0,0x0,0x0,0x0,0x4,0x0,0x1,0xee,0x0,0x4,0x4,0x4,0x4
[2023-02-03 14:18:47.73]	DEBUG:	send bulk cmd 0x1,0x40,0xf4,0x0,0x0,0x0,0x0,0x4,0x0,0x1,0xee,0x0,0x5,0x5,0x5,0x5
[2023-02-03 14:18:47.81]	DEBUG:	send bulk cmd 0x1,0x40,0xf4,0x0,0x0,0x0,0x0,0x4,0x0,0x1,0xee,0x0,0x6,0x6,0x6,0x6
[2023-02-03 14:18:47.89]	DEBUG:	send bulk cmd 0x1,0x40,0xf4,0x0,0x0,0x0,0x0,0x4,0x0,0x1,0xee,0x0,0x7,0x7,0x7,0x7
[2023-02-03 14:18:47.99]	DEBUG:	send bulk cmd 0x1,0x40,0xf4,0x0,0x0,0x0,0x0,0x4,0x0,0x1,0xee,0x0,0x8,0x8,0x8,0x8
[2023-02-03 14:18:47.114]	DEBUG:	send bulk cmd 0x1,0x40,0xf4,0x0,0x0,0x0,0x0,0x4,0x0,0x1,0xee,0x0,0x9,0x9,0x9,0x9

- Example METERLiNK packet log

[illegible]



3.4.2 Device Bulk Command

Command from meter to app to describe how to transfer data from meter to app. All the **endianness** (the order or sequence of byte) is **Big-endian**

3.4.2.1 Device Bulk Info Command

- ML_DEV_BULK_INFO**

Command from meter to App. Once meter get "Ready" signal from App, meter should send "Bulk Info" to App to describe all the relative information regarding the file which would like to be transferred.

OP	Sub OP	Pad	Total Files count	Curr. File index	File type
1 byte	1 byte	1 byte	1 byte	1 byte	1 byte

File Name	Maximum Pkt Size	Total File Size	Total Packet No.	CRC
48 bytes	2 bytes	4 bytes	2 bytes	2 bytes

OP	0x02, Bulk command
Sub OP	0x01, Bulk Info command
Pad	0x00, Memory aligned
Total Files count	Multiple transferring, to indicate how many files will be transferred
Curr. File index	Multiple transferring, to indicate the sequence of numbers in batch file list.
File Type	To indicate the file format

0x00, Unknow file
0x01, Image file
0x02, Video file
0x03, Log file, file with fixed packet size for transmission, and might be smaller in the last packet. (refer max pkt. Size, total file size, total packet size)
0x04, TBD
0x05, Dynamic log, file contents same measurement unit. Using different packet size for transmission. (refer max pkt.=0, total file size, total)
0x06, EVSE Tester Serial log

File Name	File name and subfile name, ex: IMG0001.jpg
-----------	---

Maximum Packet Size	- To describe the maximum ML_DEV_BULK_DATA data length. The maximum size of the one packet size should depend on MTU Size, for more detail please check "BLE METERLiNK Host SPI-bus Commands 1.XX.pdf". - Default packet Size should less then 180 Bytes. To increase throughput developer can get MTU size by SPI command to determine the maximum packet size - Maximum Packet Size should be multiple of 4. - Regarding Dynamic log (0x05) Please check ML_DEV_BULK_DATA example
---------------------	---

Total File Size	To indicate how many byte will be transferred regarding this file Total Packet No. = 100, Maximum Packet size = 230, Total File Size <= 23000 Byte (the last packet is not full of Maximum packet size)
-----------------	--

Total Packet No.	To indicate how many packets will be transferred regarding this file
CRC	Using the CRC-16/CCITT-FALSE, calculating data from OP code field to Total Packet No.



- **ML_DEV_BULK_INFO_REPLY**

Reply command from app to meter. If there is no error happen, App won't need to reply. Only errors happened. Apps should reply to errors to meter immediately. Meter will do corresponding behavior.

OP	Sub OP	Rsp	Pad
1	1	1	1

OP	0x02
Sub OP	0x01
Rsp	Response code, Check response code table
Pad	0x00

3.4.2.2 Device Bulk Data Command

- **ML_DEV_BULK_DATA**

Command from meter to App. Once, "Bulk Info" has sent. Meter should separate the file to small packet as following format to send to App separately.

- **The Total packet size should depend on MTU size. Default packet Size should be less than 180 Bytes.**
- **To increase throughput developer can get MTU size by SPI command to determine the maximum packet size**

OP	Sub OP	Pad1	Curr. File Index	Curr Packet Index	Actual Data size
1	1	1	1	2	2

Data	Pad2	CRC
?	2	2

OP	0x02, Bulk Command
Sub OP	0x02, Bulk Data command
Pad1	0x00, Memory aligned
Curr. File index	For multiple file selection feature to indicate this packet belongs to which file. Index start counting from 1. (If only single file would like to transfer, this field should be 1)
Curr. Packet Index	The sequence number of this packet, Index starts counts from 1 To indicate how much data in this packet, the value must be less or equal than Maximum Packet Size in ML_DEV_BULK_INFO

Actual Data Size

- For Driver developer, Actual Data Size should be dependent on MTU Size, for more detail please check "**BLE METERLiNK Host SPI-bus Commands 1.XX.pdf**".
- Actual Data Size field should be less than "MTU maximum size" and The length should be multiple of 4.
- **The total size of ML_DEV_BULK_DATA (from OP to CRC) must be less than MTU size, Ex: MTU size: 180 Byte,**



Data = 180 – 4 Byte (OP to curr. File index) – 4 Byte (Curr Packet Index,
Actual Data size) – 4 Byte (Pad2, CRC)

Data The data size should be same as bulk info “**Actual Data Size**”
Pad1 Memory aligned
CRC Using the CRC-16/CCITT-FALSE, calculating data from OP code field to Pad2

Example: dynamic log (0x05) regarding Bulk data’s Data field:

Assume **Bulk info’s Maximum Packet Size** = 8 byte

And, if meter would like to send 5-byte data only, the **Bulk data’s Data Size** = 5

For example, meter would like to send “0x11 0x22 0x33 0x44 0x55”

Therefore, **Bulk data’s Data field** should be 8 bytes, including 3-byte 0x00 dummy data as follows:

“0x11 0x22 0x33 0x44 0x55 0x00 0x00 0x00”

- Meter should assign data and append dummy data (0x00) to equal Maximum Packet size
- App should check the Data size filed to check the correct data length.

• ML_DEV_BULK_DATA_REPLY

Reply command from app to meter. If there is no error happen, App won’t need to reply. Only errors happened. App should reply to errors to meter immediately. Meter will do corresponding behavior.

OP	Sub OP	Rsp	Pad
1	1	1	1

OP 0x02
Sub OP 0x02
Rsp Response code, [Check response code table](#)
Pad 0x00

3.4.2.3 Device Bulk Cancel Command

• ML_DEV_BULK_CANCEL

Command from meter to App to cancel file transferring, Once App gets this command, it will terminate file transferring and indicate file transferring is canceled.

OP	Sub OP	Pad	Pad
1	1	1	1

OP 0x02, Bulk Command



Sub OP 0x03, Bulk Cancel command to cancel bulk data transferring
Pad1 0x00, NULL
Pad2 0x00, NULL

3.4.5 Device Message to App Command

Command from meter to app, to asking app information or status.

3.4.5.1 Device to Get App Busy Status Command

- ML_DEV_MSG_GET_APP_BUSY_STATUS**

Meter send GET_APP_BUSY_STATUS command to ask app status (Busy or Ready). Meter should wait for a response from app.

OP	Sub OP	Pad	Pad
1	1	1	1

OP 0x05
Sub OP 0x01, Asking does app ready to receive data
Pad 0x00
Pad 0x00

- ML_DEV_MSG_GET_APP_BUSY_STATUS_REPLY**

Reply command from app to meter. GET_APP_BUSY_STATUS_REPLY command describes status of app (Busy or Ready). If app is in busy state, meter should wait a moment and send GET_APP_BUSY_STATUS again.

OP	Sub OP	Rsp	Pad
1	1	1	1

OP 0x05
Sub OP 0x01
Rsp Response code, reply BUSY or READY to receive packet from meter, [Check response code table](#)
Pad 0x00



3.4.5.2 Device Common Status Command

Command from meter to app to indicate status of meter

OP	Sub OP	Pad	Pad
1	1	1	1

OP	0x05
Sub OP	0xFE, BLE Rx Queue full 0xFD, BLE Rx Queue not full 0xFC, Device Storage full
Pad	0x00
Pad	0x00

3.5 METERLiNK Rx Commands

Command from App to meter. Meter will receive command via "METERLiNK RX Characteristic"

5300 2010 0046 4C49 522D 54414957414E	(C) METERLiNK RX Characteristic
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3.5.1 App Engineer Command

TBD

3.5.2 App Message Control Command

• ML_APP_MSG_CONTROL

Message from App to meter to control BLE and Meter. Using ML_APP_MSG_CONTROL to Start/Stop sensor reading and bulk transferring

OP	Sub OP	Pad	Pad
1	1	1	1

OP	0xFD
Sub OP code	0xFE, start sensor reading (Not supported) 0xFD, stop sensor reading packet (Not supported) 0xFC, start bulk transferring (Not supported) 0xFB, stop bulk transferring, (Support this command)
Pad	0x00
Pad	0x00



• ML_APP_MSG_CONTROL_REPLY

Message from meter to App to reply command is success or fail.

OP	Sub OP	Rsp	Pad
1	1	1	1

OP 0xFD

Sub OP 0xFE, start sensor reading (Not supported)
0xFD, stop sensor reading packet (Not supported)
0xFC, start bulk transferring (Not supported)
0xFB, stop bulk transferring (Currently support this command only)
0xFA, BLE will do OTA upgrade. Meter should STOP communicate with BLE
(Not supported now)

Rsp Check Responsed code table

3.5.3 App Camera & IGM Control Command

App message to meter to control camera and IGM.

4.0 OTA

APP support needed: Firmware Over-The-Air upgrade. (DFU in BLE solution term)

Reference:

OTA GitHub Source base and related info: (We are using Nordic solution)

- iOS: <https://github.com/NordicSemiconductor/IOS-nRF-Toolbox>
 - dependency1: cocoapod
 - dependency2: DFU lib: <https://github.com/NordicSemiconductor/IOS-Pods-DFU-Library>
 - The above items were verified with my Macbook and compiled passed.
- Android: <https://github.com/NordicSemiconductor/IOS-nRF-Toolbox>
 - Dependency: Android BLE lib: <https://github.com/NordicSemiconductor/Android-BLE-Library/>
- Nordic official supports: (We are using nRF5 SDK v17.0.2)
 - SDK Doc: DFU(Device Firmware Update) process:
https://infocenter.nordicsemi.com/topic/sdk_nrf5_v17.0.2/lib_bootloader_dfu_process.html
 - Dev forum: <https://devzone.nordicsemi.com/>

4.1 Signed image deployment

(Under construction) Flow Method description



5.0 Meter Type Index

Sensor Type IDs	Functionality Type	Display in App
0x00	Other	Other
0x01	DMM	Digital Multimeter
0x02	Clamp	Clamp
0x03	Temperature	Temperature
0x04	Relative Humidity	Relative Humidity
0x05	Anemometer	Anemometer
0x06	Manometer	Manometer
0x07	Moisture	Moisture
0x08	Gas Sensor	Gas Sensor
0x09	Water Quality	Water Quality
0x0A	Vibration	Vibration
0x0B	Sound	Sound
0x0C	Weight Scale	Weight Scale
0x0D	Speed	Speed
0x0E	Tachometer	Tachometer
0x0F	Pressure	Pressure
0x10	Irradiance	Irradiance
0x11	Borescope	Borescope
0x12	Open Jaw	Open Jaw
0x13	Environment	Environment
0x14	Insulation	Insulation
0x15	Voltage Detector	Voltage Detector
0x16	Light Meter	Light
0x17	Temperature & Humidity meter	Temperature & Humidity
...		
...		
0XE0	Vibration Sensor	Vibration Sensor
...		
...		
0XF0	EVSE Tester	EVSE Tester
(up to 0xFD)		
0xFF	(Un-allowed)	(Un-allowed)



5.1 Function Name Index

Index	Value	Alt Reading Format	Display in App	Log display name
0x00	None		(No function, no show)	(No function, no show)
0x01	Other		Other	Other
0x02	Magnet Detector	binary	Magnet Detector	Magnet Detector
0x03	NCV	binary	NCV	NCV
0x04	Battery Test		Battery Test	Battery Test
0x05	Diode Test		Diode Test	Diode Test
0x06	Smart Diode		Smart Diode	Smart Diode
0x07	Continuity	binary	Continuity	Continuity
0x08	DC Voltage		DC V	DC V
0x09	AC Voltage		AC V	AC V
0x0A	DC Current		DC Amp.	DC Amp.
0x0B	DC Current - flex		DC Amp. flex	DC Amp. flex
0x0C	DC Current - clamp		DC Amp. clamp	DC Amp. clamp
0x0D	DC Current - clamp tip		DC Amp. clamp tip	DC Amp. clamp tip
0x0E	AC Current		AC Amp.	AC Amp.
0x0F	AC Current - flex		AC Amp. flex	AC Amp. flex
0x10	AC Current - clamp		AC Amp. clamp	AC Amp. clamp
0x11	AC Current - clamp tip		AC Amp. clamp tip	AC Amp. clamp tip
0x12	Conductance		Conductance	Conductance
0x13	Admittance		Admittance	Admittance
0x14	Impedance		Impedance	Impedance
0x15	Resistance		Resistance	Resistance
0x16	Capacitance		Capacitance	Capacitance
0x17	Inductance		Inductance	Inductance
0x18	Frequency		Freq.	Freq.
0x19	RPM		RPM	RPM
0x1A	Duty Cycle		Duty Cycle	Duty Cycle
0x1B	Test Voltage		Test Volt.	Test Volt.
0x1C	Insulation Test Voltage		Insulation Test Volt.	Insulation Test Volt
0x1D	Insulation Resistance		Insulation Resis.	Insulation Resis
0x1E	Polarization Index (PI)		PI	PI
0x1F	Dielectric absorption ratio (DAR)		DAR	DAR



0x20	Earth bond Test Voltage		Earth bond Test Volt.	Earth bond Test Volt.
0x21	Earth bond		Earth bond	Earth bond
0x22	4-20mA %		4-20mA %	4-20mA
0x23	DC Power		DC Power	DC Power
0x24	Active Power		Active Power	Active Power
0x25	Reactive Power		Reactive Power	Reactive Power
0x26	Apparent Power		Apparent Power	Apparent Power
0x27	Energy		Energy	Energy
0x28	Active Power Energy		Active Power Energy	Active Power Energy
0x29	Reactive Power Energy		Reactive Power Energy	Reactive Power Energy
0x2A	Apparent Power Energy		Apparent Power Energy	Apparent Power Energy
0x2B	Power Factor		PF	PF
0x2C	Phase		Phase	Phase
0x2D	Harmonic level		Harmonic level	Harmonic level
0x2E	Total Harmonic Distortion (THD)		THD	THD
0x2F	Crest factor		Crest factor	Crest factor
0x30	Quality factor		Quality factor	Quality factor
0x31	Dissipation factor		Dissipation factor	Dissipation factor
0x32	ESR		ESR	ESR
0x33	AC Load Tester Load		AC Load Tester	AC Load Tester
0x34	AC Load Tester Unloaded Voltage		AC Unloaded Volt.	AC Unloaded Volt
0x35	AC Load Tester Loaded Voltage		AC Loaded Volt.	AC Loaded Volt
0x36	AC Load Tester Voltage Drop		AC Volt. Drop	AC Volt Drop
0x37	Receptacle Tester	Receptacle Tester Code	Receptacle Tester	Receptacle Tester
0x38	AFCI	Receptacle Tester Code	AFCI	AFCI
0x39	GFCI	Receptacle Tester Code	GFCI	GFCI
	* Use "Index number"	0x00 No Result	No Result	No Result
	*	0x01 Correct	Correct	Correct
	*	0x02 H-N reversed	H-N reversed	H-N reversed
	*	0x03 No Ground	No Ground	No Ground
	*	0x04 H-N rev, no Gnd	H-N rev, no Gnd	H-N rev, no Gnd
	*	0x05 Open Hot	Open Hot	Open Hot
	*	0x06 Open Neutral	Open Neutral	Open Neutral
	*	0x07 H-G Reversed	H-G Reversed	H-G Reversed
	* Use "Index number"	0x08 Other Failure	Other Failure	Other Failure
0x3A	GFCI Trip Current		GFCI Trip Amp.	GFCI Trip Amp.
0x3B	GFCI Trip Time		GFCI Trip Time	GFCI Trip Time
0x3C	EPD Trip Current		EPD Trip Amp.	EPD Trip Amp.
0x3D	Neutral-Ground voltage		Neutral-Ground Volt.	Neutral-Ground Volt
0x3E	Hot impedance		Hot impedance	Hot impedance
0x3F	Neutral impedance		Neutral impedance	Neutral impedance



Index	Value	Alt Reading Format	Display in App	Log display name
0x40	Ground impedance		Ground impedance	Ground impedance
0x41	Available Short Circuit Current		Available Short Circuit Amp.	Available Short Circuit Amp.
0x42	Emissivity		Emissivity	Emissivity
0x43	Temperature		Temp.	Temp.
0x44	Temperature - air		Temp. Air	Temp. Air
0x45	Temperature - heat index		Temp. Heat index	Temp. Heat index
0x46	Temperature - wind chill		Temp. Wind chill	Temp. Wind chill
0x47	Temperature - dry bulb (DBT)		Temp. DBT	Temp. DBT
0x48	Temperature - wet bulb (WBT)		Temp. WBT	Temp. WBT
0x49	Temperature - WGBT		Temp. WGBT	Temp. WGBT
0x4A	Temperature - TG		Temp. TG	Temp. TG
0x4B	Temperature - surface/IR		Temp. Surface/IR	Temp. Surface/IR
0x4C	Temperature - thermocouple		Temp. - TC	Temp TC
0x4D	Temperature - thermocouple type	Thermocouple Type Code	Temp. - TC type	Temp TC type
	* Use "Index number"	0x00: Unknown	TC Unknown	TC Unknown
	*	0x01: E	TC E	TC E
	*	0x02: J	TC J	TC J
	*	0x03: K	TC K	TCK
	*	0x04: M	TC M	TC M
	*	0x05: N	TC N	TC N
	*	0x06: T	TC T	TCT
	*	0x07: B	TC B	TC B
	*	0x08: R	TC R	TC R
	*	0x09: S	TC S	TC S
	*	0x0A: C	TC C	TC C
	*	0x0B: D	TC D	TC D
	*	0x0C: G	TC G	TC G
	*	0x0D: Chromel	TC Chromel	TC Chromel
	*	0x0E: P	TC P	TC P
	*	0x0F: Pt/Mo	TC Pt/Mo	TC Pt/Mo
	*	0x10: Ir/Rh	TC Ir/Rh	TC Ir/Rh
	*	0x11: Au/Pt	TC Au/Pt	TC Au/Pt
	*	0x12: Pt/Pd	TC Pt/Pd	TC Pt/Pd
	* Use "Index number"	0x13: Other	TC Other	TC Other
0x4E	Relative Humidity		RH	RH
0x4F	Air Pressure		Air Pressure	Air Pressure
0x50	Dew Point		DEW	DEW
0x51	Mixing Ratio		MIX	MIX
0x52	Vapor Pressure		VAP	VAP
0x53	Air Velocity		Air Velocity	Air Velocity
0x54	Beaufort force		Beaufort force	Beaufort force
0x55	Air Flow		Air Flow	Air Flow



Index	Value	Alt Reading Format	Display in App	Log display name
0x56	CO concentration		CO Conc.	CO Conc
0x57	HCHO concentration		HCHO Conc.	HCHO Conc
0x58	CH ₂ O concentration		CH ₂ O Conc.	CH ₂ O Conc
0x59	CO ₂ concentration		CO ₂ Conc.	CO ₂ Conc
0x5A	Refrigerant Leakage		Refrigerant Leakage	Refrigerant Leakage
0x5B	Combustible Gas		Combustible Gas	Combustible Gas
0x5C	Particle Size		Particle Size	Particle Size
0x5D	Particle Count		Particle Count	Particle Count
0x5E	Light level		Light level	Light level
0x5F	UV level		UV level	UV level
0x60	Sound Level		Sound level	Sound level
0x61	Moisture		Moist.	Moist.
0x62	Moisture - resistive		Moist. - resistive	Moist resistive
0x63	Moisture material type		Moist. material type	Moist material type
0x64	Moisture - capacitive		Moist. - capacitive	Moist capacitive
0x65	Moisture - soil		Moist. - soil	Moist soil
0x66	Time		Time	Time
0x67	Distance		Distance	Distance
0x68	Area		Area	Area
0x69	Volume		Volume	Volume
0x6A	EMF frequency		EMF Freq.	EMF Freq
0x6B	EMF strength		EMF strength	EMF strength
0x6C	Acceleration		Accel	Accel
0x6D	Acceleration - X		Accel X	Accel X
0x6E	Acceleration - Y		Accel Y	Accel Y
0x6F	Acceleration - Z		Accel Z	Accel Z
0x70	Weight		Weight	Weight
0x71	Acidity		Acidity	Acidity
0x72	ORP		ORP	ORP
0x73	TDS		TDS	TDS
0x74	Salinity		Salinity	Salinity
0x75	Dissolved oxygen		Dissolved oxygen	Dissolved oxygen
0x76	Chlorine concentration		Chlorine Conc.	Chlorine Conc
0x77	Turbidity		Turbidity	Turbidity
0x78	Fluoride concentration		Fluoride Conc.	Fluoride Conc
0x79	DC Voltage – LoZ	Related to 0x96	DC Volt. LoZ	DC Volt LoZ
0x7A	AC Voltage – LoZ		AC Volt. LoZ	AC Volt LoZ
0x7B	AC Voltage – VFD		AC Volt. VFD	AC Volt VFD
0x7C	AC Voltage – LoZ with VFD		AC Volt. LoZ with VFD	AC Volt LoZ with VFD
0x7D	AC Current – VFD		AC Amp. VFD	AC Amp. VFD
0x7E	AC Current – Flex with VFD		AC Amp. Flex with VFD	AC Amp. Flex with VFD
0x7F	AC Current – Clamp with VFD		AC Amp. Clamp with VFD	AC Amp. Clamp with VFD
0x80	Irradiance		Irradiance	Irradiance



0x81 Index	Compass Bearing Value	Alt Reading Format	Compass Bearing Display in App	Compass Bearing Log display name
0x82	Pitch – inclination, slope, tilt		Pitch	Pitch
0x83	Latitude		Latitude	Latitude
0x84	Longitude		Longitude	Longitude
0x85	Altitude		Altitude	Altitude
0x86	Count		Count	Count
0x87	IGM – Spot Temperature		IGM Spot Temp.	IGM Spot Temp.
0x88	IGM – ROI High Temperature		IGM ROI High Temp.	IGM ROI High Temp.
0x89	IGM – ROI Low Temperature		IGM ROI Low Temp.	IGM ROI Low Temp.
0x8A	AC+DC Voltage		AC+DC Volt.	AC+DC Volt.
0x8B	AC+DC Current		AC+DC Curr.	AC+DC Curr.
0x8C	AC+DC Current-Clamp		AC+DC Curr. Clamp	AC+DC Curr. Clamp
0x8D	Watt		Watt	Watt
0x8E	Photovoltaic DC	0x00	PV DC	PV DC
	* Use “Index number”	0x01	PV DC V	PV DC V
	*	0x02	PV DC A	PV DC A
	*	0x03	PV DC Watt	PV DC Watt
	* Use “Index number”	0xFF	n/a	n/a
0x8F	Photovoltaic DC Watt		PV Watt	PV Watt
0x90	Lepton Flux	Unit: Flux	Lepton Flux	Lepton Flux
0x91	Lepton Telemetry FPA	Unit: temperature °C	Lepton T FPA	Lepton T FPA
0x92	Lepton I2C FPA	Unit: temperature °C	Lepton I2C FPA	Lepton I2C FPA
0x93	Insulation	0x00	Insulation	Insulation
	* Use “Index number”	0x01	INSUL. RES. 50V	INSUL. RES. 50V
	*	0x02	INSUL. RES. 100V	INSUL. RES. 100V
	*	0x03	INSUL. RES. 250V	INSUL. RES. 250V
	*	0x04	INSUL. RES. 500V	INSUL. RES. 500V
	*	0x05	INSUL. RES. 1KV	INSUL. RES. 1KV
	*	0x06	INSUL. RES. 2.5KV	INSUL. RES. 2.5KV
	*	0x07	INSUL. RES. 5KV	INSUL. RES. 5KV
	*	0x20	PI	PI
	*	0x21	INSUL. RES. 50V, PI	INSUL. RES. 50V, PI
	*	0x22	INSUL. RES. 100V, PI	INSUL. RES. 100V, PI
	*	0x23	INSUL. RES. 250V, PI	INSUL. RES. 250V, PI
	*	0x24	INSUL. RES. 500V, PI	INSUL. RES. 500V, PI
	*	0x25	INSUL. RES. 1KV, PI	INSUL. RES. 1KV, PI
	*	0x26	INSUL. RES. 2.5KV, PI	INSUL. RES. 2.5KV, PI
	*	0x27	INSUL. RES. 5KV, PI	INSUL. RES. 5KV, PI
	*	0x30	DAR	DAR
	*	0x31	INSUL. RES. 50V, DAR	INSUL. RES. 50V, DAR
	*	0x32	INSUL. RES. 100V, DAR	INSUL. RES. 100V, DAR
	*	0x33	INSUL. RES. 250V, DAR	INSUL. RES. 250V, DAR
	*	0x34	INSUL. RES. 500V, DAR	INSUL. RES. 500V, DAR



	*	0x35	INSUL. RES. 1KV, DAR	INSUL. RES. 1KV, DAR
	*	0x36	INSUL. RES. 2.5KV, DAR	INSUL. RES. 2.5KV, DAR
	* Use "Index number"	0x37	INSUL. RES. 5KV, DAR	INSUL. RES. 5KV, DAR
0x94	Period		Peroid	Period
0x95	Decibel	0x00	dB	dB
	* Use "Index number"	0x01	dBm	dBm
	* Use "Index number"	0xFF	n/a	n/a
0x96	AC+DC Volt. LoZ	Related to 0x79	AC+DC Volt. LoZ	AC+DC Volt. LoZ
0x97	CP State		CP State	CP State
0x98	Inrush		Inrush	Inrush
0x99	Displacement (Peak to Peak)		DISPL P-P	DISPL P-P
0x9A	Kurtosis		Kurtosis	Kurtosis
0x9B	Skewness		Skewness	Skewness
0x9C	Deviation		Deviation	Deviation
0x9D	Acceleration RMS		Accel RMS	Accel RMS
0x9E	Acceleration Mean		Accel Mean	Accel Mean
0x9F	Acceleration Median		Accel Median	Accel Median
0XA0	Acceleration (0 to Peak)		Accel 0-P	Accel 0-P
0xA1	OA X		OA X	OA X
0XA2	OA Y		OA Y	OA Y
0xA3	OA Z		OA Z	OA Z



5.2 Measurement Unit Index

Index	Value	Sub Unit Index	Display in App	Log display name
0x00	unit-less		(No unit, no show)	(No unit, no show)
	Binary – Detected/Not Detected	0x00 (Default)	n/a	n/a
0x01	* Subunits index	0x01	Detected	Detected
	* Subunits index	0x02	Not Detected	Not Detected
0x02	Volt		V	V
0x03	Amp		A	A
0x04	Ohm		Ohm	Ohm
0x05	Siemen		S	S
0x06	Farad		F	F
0x07	Henry		H	H
0x08	Hz		Hz	Hz
0x09	RPM		RPM	RPM
0x0A	%		%	%
0x0B	Watt		W	W
0x0C	VA		VA	VA
0x0D	Joule		Joule	Joule
0x0E	kWh		kWh	kWh
0x0F	mVAh		mVAh	mVAh
0x10	degree - thermal		degree	degree
0x11	dB	0x00 (Default)	dB	dB
	* Subunits index	0x01	dBm	dBm
	*	0x02	dB(A)	dB(A)
	*	0x03	dB(B)	dB(B)
	* Subunits index	0x04	dB(C)	dB(C)
0x12	RESERVED		RESERVED	RESERVED
0x13	second		second	second
0x14	°C		°C	Degree C
0x15	°F		°F	Degree F
0x16	K		K	K
0x17	%RH		%RH	%RH
0x18	PSI		psi	psi
0x19	Pa		Pa	Pa
0x1A	bar		bar	bar
0x1B	inHg		inHg	inHg
0x1C	mmHg		mmHg	mmHg
0x1D	GPP		GPP	GPP
0x1E	g/kg		g/kg	g/kg
0x1F	ft/min		ft/min	ft/min
0x20	m/s		m/s	m/s
0x21	km/h		km/h	km/h
0x22	mph		Mph	Mph



0x23	knots		Knots	Knots
0x24	ft/s ²		ft/s ²	ft/s ²
0x25	m/s ²		m/s ²	m/s ²
0x26	Gravity		G	G
0x27	CFM		CFM	CFM
0x28	CMM		CMM	CMM
0x29	PPT		PPT	PPT
0x2A	PPM		PPM	PPM
0x2B	oz/year		oz/year	oz/year
0x2C	% LEL		%LEL	%LEL
0x2D	um		um	um
0x2E	RESERVED		RESERVED	RESERVED
0x2F	W/cm ²		W/cm ²	W/cm ²
0x30	RESERVED		RESERVED	RESERVED
0x31	% (by wt)		%(by wt)	%(by wt)
0x32	%WME		%WME	%WME
0x33	second		sec.	sec.
0x34	min		min.	min.
0x35	hour		hour	hour
0x36	M		m	m
0x37	m ²		m ²	m ²
0x38	m ³		m ³	m ³
0x39	In		in	in
0x3A	in ²		in ²	in ²
0x3B	in ³		in ³	in ³
0x3C	Ft		ft	ft
0x3D	ft ²		ft ²	ft ²
0x3E	ft ³		ft ³	ft ³
0x3F	V/m		V/m	V/m
0x40	A/m		A/m	A/m
0x41	W/m ²		W/m ²	W/m ²
0x42	Tesla		Tesla	Tesla
0x43	Gauss		Gs	Gs
0x44	gram		g	g
0x45	oz		oz	oz
0x46	lbs		lbs	lbs
0x47	pH		pH	pH
0x48	g/l		g/l	g/l
0x49	Liter		L	L
0x4A	NTU		NTU	NTU
0x4B	Binary – Pass (1)/Fail (0)	0x00 (Default)		
	* Subunits index	0x01	Pass	Pass
	* Subunits index	0x02	Fail	Fail
0x4C	Binary – On/Off			



	* Subunits index	0x01	On	On
	* Subunits index	0x02	Off	Off
0x4D	degree – angle		degree - angle	degree - angle
0x4E	Count		count	count
0x4F	Fc		Fc	Fc
0x50	Lux		Lux	Lux
0x51	Power Factor		PF	PF
0x52	Flux		Flux	Flux
0x53	Ratio		Ratio	Ratio
0x54	degree		degree	degree
0x55	Gravity		g	g
0x56	mm/s RMS		mm/s RMS	mm/s
0x57	inch/s RMS		inch/s RMS	inch/s RMS



5.3 Vibration ISO Index

ISO	ISO name	Alert mm/s (inch/s)	Danger mm/s (inch/s)	Note
0x00	NA	NA	NA	NA
0x01	Large Machines / Rigid $P > 300 \text{ kW}$, $H \geq 315 \text{ mm}$	4.5 (0.18)	7.1 (0.28)	ISO10816-3 Large Equip.
0x02	Large Machines / Flexible $P > 300 \text{ kW}$, $H \geq 315 \text{ mm}$	7.1 (0.28)	11 (0.43)	ISO10816-3 Large Equip.
0x03	Medium Machines / Rigid $15 \text{ kW} < P \leq 300 \text{ kW}$, $160 \text{ mm} \leq H \leq 315 \text{ mm}$	2.8 (0.11)	4.5 (0.18)	ISO10816-3 Large Equip.
0x04	Medium Machines / Rigid $15 \text{ kW} < P \leq 300 \text{ kW}$, $160 \text{ mm} \leq H \leq 315 \text{ mm}$	4.5 (0.18)	7.1 (0.28)	ISO10816-3 Large Equip.
...				
0xFF	End of IOS index			

3. ISO 10816-3 standard					
Vibration speed: 10-1000 Hz $r > 600 \text{ rpm}$, 2-1000 Hz $r > 120 \text{ rpm}$					
Machine type		Medium-sized equipment $15 \text{ kW} < P \leq 300 \text{ kW}$		Large equipment $300 \text{ kW} < P < 50 \text{ MW}$	
		motor $160 \text{ mm} \leq H < 315 \text{ mm}$		motor $315 \text{ mm} \leq H$	
basis		rigid	Flexible	rigid	Flexible
Inch/s rms	mm/s rms				
0.03	0.71	A			
0.06	1.4				
0.09	2.3	B			
0.11	2.8				
0.14	3.5	C			
0.18	4.5				
0.28	7.1	D			
0.43	11				

Vibration sensor ISO 10816-3	Alert mm/s (inch/s)	Danger mm/s (inch/s)
Large Machines / Rigid $P > 300 \text{ kW}$, $H \geq 315 \text{ mm}$	4.5 (0.18)	7.1 (0.28)
Large Machines / Flexible $P > 300 \text{ kW}$, $H \geq 315 \text{ mm}$	7.1 (0.28)	11 (0.43)
Medium Machines / Rigid $15 \text{ kW} < P \leq 300 \text{ kW}$, $160 \text{ mm} \leq H \leq 315 \text{ mm}$	2.8 (0.11)	4.5 (0.18)
Medium Machines / Flexible $15 \text{ kW} < P \leq 300 \text{ kW}$, $160 \text{ mm} \leq H \leq 315 \text{ mm}$	4.5 (0.18)	7.1 (0.28)



6.0 Meter Log Format Example

6.0.1 Sampling rate and Sampling Unit (unit is Second)

case1: "3, 1," -> In 1s have 3 data, the sampling rate will be 0.333 (3 digit)

case2: "1, 60" -> In 60s has 1 data "

6.0.2 Device without timestamp

Every field in log file should be fill the correct value. **In the case which the device without timestamp does not fill in anything in this filed.**

6.0.3 Example of "Value" data format

- Value can present as "9.0" or "9", both is acceptable
- While OL happened, the value should be fill correct "Value" with "Power of 10"

6.0.4 Example of "Power of 10"

Most of "power of 10" will be (9, 6, 3, 0, -3, -6, -9)

6.0.5 Out of Range flag and value

- If the value is "out of range" this filed should be "1" to indicate this value is "out of range". And the "value" filed should depends on OL Max/Min value
 - **Example1:** If OL max is 999. the out of value should be 999 or more than 999 (ex: 1000). And the "Out of Rang" flag should be 1 to indicate "out of range"
 - **Example2:** If OL min is 0.999, the out of value should be 0.999 or less than 0.999 (ex: 0.0001). And the "Out of Rang" flag should be 1 to indicate "out of range"
 - **Example3:** OL min is -999 the out of value will be -999 or less than -999 (ex: -1000). And the "Out of Rang" should be 1 to indicate "out of range"

Note:

- "If AC = RMS value "the OL Min will always to 0
- AC and Resistor don't have negative values. OL flag doesn't need to be asserted



6.1 Log CSV Format

	A	B	C	D	E	F	G	H	I
1	Device Model	Function Name	Function Index	Unit Name	Unit Index	OL Max	OL Min	Sampling Rate	Sampling Unit (second)
2	FLIR-DM286 (SN12345678)	DC Volt	0x09	Volt	0x02	999999	0.999999	3	1
3									
4	NO	DATE	TIME	VALUE	POWER OF 10	UNIT	OUT OF RANGE		
5	0	6/29/2022	14:04:34	10.01	3	Volt			
6	1	6/29/2022	14:04:35	9	6	Volt			
7	2	6/29/2022	14:04:35	1	7	Volt	1		
8	3	6/29/2022	14:04:36	9.9	-3	Volt			
9	4	6/29/2022	14:04:36	9	-6	Volt			
10	5	6/29/2022	14:04:37	9	-7	Volt	1		
11	6	6/29/2022	14:04:37	7	3	Volt			
12	7	6/29/2022	14:04:38	5	0	Volt			
13	8	6/29/2022	14:04:38	5	3	Volt			
14	9	6/29/2022	14:04:39	0.7	6	Volt			
15	10	6/29/2022	14:04:39	3	3	Volt			
16	11	6/29/2022	14:04:40	4	-3	Volt			
17	12	6/29/2022	14:04:40	2	-6	Volt			
18	13	6/29/2022	14:04:41	5	-6	Volt			
19	14	6/29/2022	14:04:41	6	0	Volt			
20	15	6/29/2022	14:04:42	1	3	Volt			
21	16	6/29/2022	14:04:42	4	6	Volt			
22	17	6/29/2022	14:04:43	8	5	Volt			
23	18	6/29/2022	14:04:43	5	-3	Volt			
24									

6.2 Log Raw File Format

6.2.1 Log Header Format

Log header format should follow the rule to fill correct data. Since METERLiNK App will parse each word on the header. In each word of beginning should be upper case.

Device Model,Function Name,Function Index,Unit Name,Unit Index,OL Max,OL Min,Sampling Rate,Sampling Unit (second)
FLIR-DM286 (SN12345678),DC Volt,0x09,Volt ,0x02,999999,0.999999,3,1

6.2.2 Log Data Format

Log data format aslo need to follow below format

DATE: MM/DD/YYYY

TIME: HH:MM:SS (24H format)

NO,DATE,TIME,VALUE,POWER OF 10,UNIT,OUT OF RANGE,
0,6/29/2022,14:04:34,10.01,3,Volt,,
1,6/29/2022,14:04:35,9,6,Volt,,
2,6/29/2022,14:04:35,1,7,Volt,1,
3,6/29/2022,14:04:36,9.9,-3,Volt,,



Subject
METERLiNK 4.2 Specification

Info master
Sylar.Lan@Teledyne.com;Henry.Li@Tele
dyne.com,

SaveDate
2026-04-146

\ Title
FLIR METERLiNK 4.2Protocol

Doc No / Filename
METERLiNK 4.2

Rev Page
4.2.14 57 (58)

```
1 Device Model,Function Name,Function Index,Unit Name,Unit Index,OL Max,OL Min,Sampling Rate,Sampling Unit (second)
2 FLIR-DM286 (SN12345678),DC Volt,0x09,Volt ,0x02,999999,0.999999,3,1
3
4 NO,DATE,TIME,VALUE,POWER OF 10,UNIT,OUT OF RANGE,
5 0,6/29/2022,14:04:34,10.01,3,Volt,,
6 1,6/29/2022,14:04:35,9,6,Volt,,
7 2,6/29/2022,14:04:35,1,7,Volt,1,
8 3,6/29/2022,14:04:36,9.9,-3,Volt,,
9 4,6/29/2022,14:04:36,9,-6,Volt,,
10 5,6/29/2022,14:04:37,9,-7,Volt,1,
11 6,6/29/2022,14:04:37,7,3,Volt,,
12 7,6/29/2022,14:04:38,5,0,Volt,,
13 8,6/29/2022,14:04:38,5,3,Volt,,
14 9,6/29/2022,14:04:39,0.7,6,Volt,,
15 10,6/29/2022,14:04:39,3,3,Volt,,
16 11,6/29/2022,14:04:40,4,-3,Volt,,
17 12,6/29/2022,14:04:40,2,-6,Volt,,
18 13,6/29/2022,14:04:41,5,-6,Volt,,
19 14,6/29/2022,14:04:41,6,0,Volt,,
20 15,6/29/2022,14:04:42,1,3,Volt,,
21 16,6/29/2022,14:04:42,4,6,Volt,,
22 17,6/29/2022,14:04:43,8,5,Volt,,
23 18,6/29/2022,14:04:43,5,-3,Volt,,
```



7.0 Reponse Code Table

```
// Common error code
HPI_RSP_UNKNOWN          = 0x00, // no response message
HPI_RSP_ERR_CMD          = 0x01, // Command OK
HPI_RSP_ERR_UNSUPPORTED  = 0x02, // Bad command from host processor, command ignored
HPI_RSP_ERR_FRAMING_BYTES = 0x03, // The framing bytes SoH and EoT bytes not as expected
HPI_RSP_ERR_CRC          = 0x04, // CRC error
HPI_RSP_ERR_CMD_OPERATIOPN_FAIL = 0x05,

// Command OK
HPI_RSP_OK               = 0x10,

// Packet Error
HPI_RSP_ERR_PKT_LOST      = 0x30,
HPI_RSP_ERR_SEQUENTIAL_PKT_LOST = 0x31,

// BLE Staus code
HPI_RSP_BLE_ERR_CONNTION  = 0x40,
HPI_RSP_BLE_ERR_WEAK_SGINAL = 0x41,
HPI_RSP_BLE_ERR_RX_QUEUE_FULL = 0x42,
HPI_RSP_BLE_CONN_UNKNOW   = 0x43,
HPI_RSP_BLE_CONN_STANDBY  = 0x44,
HPI_RSP_BLE_CONN_ADVERTISING = 0x45,
HPI_RSP_BLE_CONN_CONNETED = 0x46,

// Status response code
HPI_RSP_STATUS_READY      = 0x50,
HPI_RSP_STATUS_BUSY       = 0x51,
HPI_RSP_STATUS_NO_PERMISSION = 0x52,
HPI_RSP_STATUS_STORAGE_FULL = 0x53,
HPI_RSP_STATUS_LOW_BATTERY = 0x54,
```